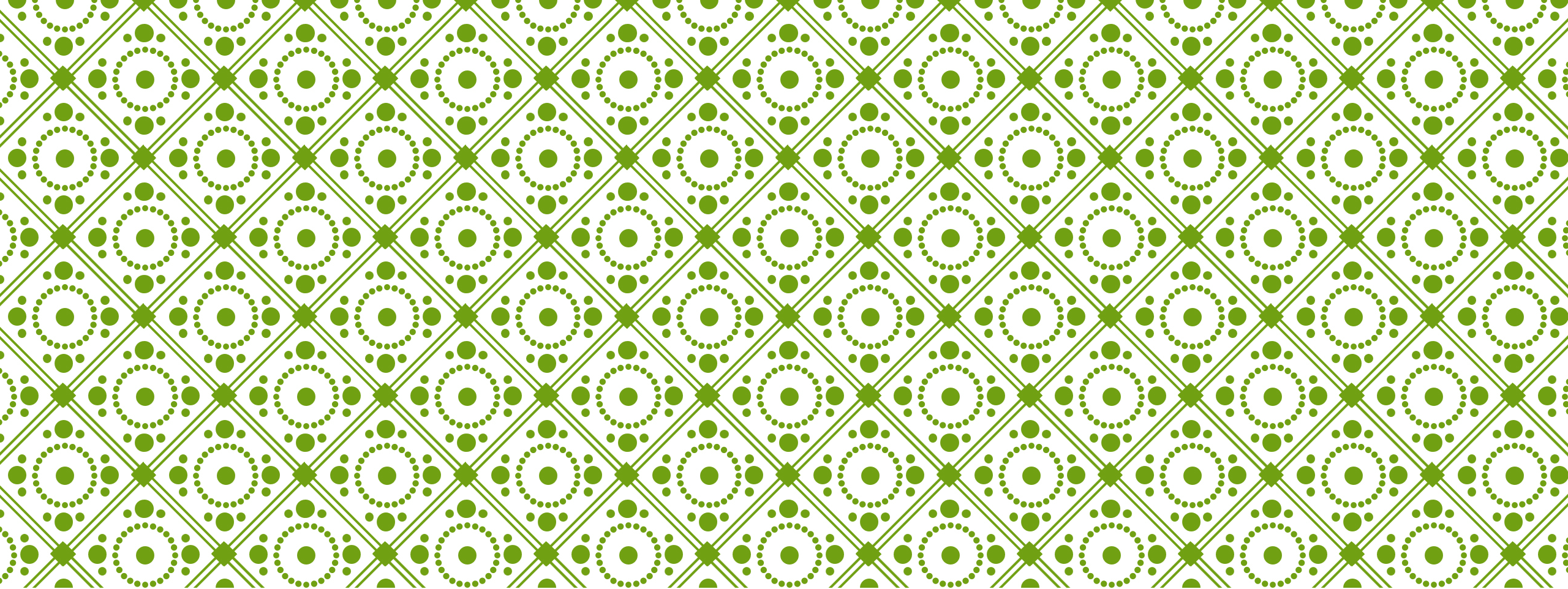




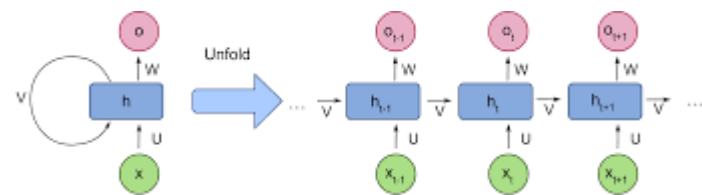
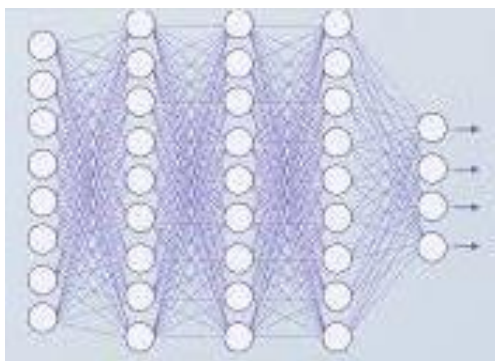
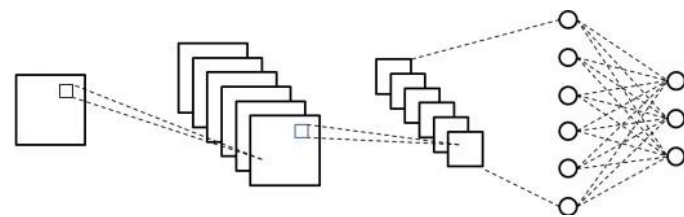
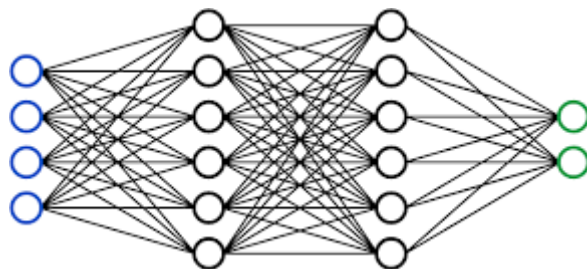
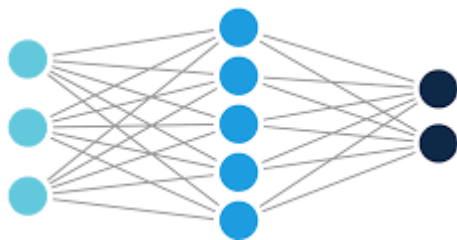
AI2002 – ARTIFICIAL INTELLIGENCE

SPRING 2024 - LECTURE 43-45

Presented By: Mr. Sandesh
Kumar

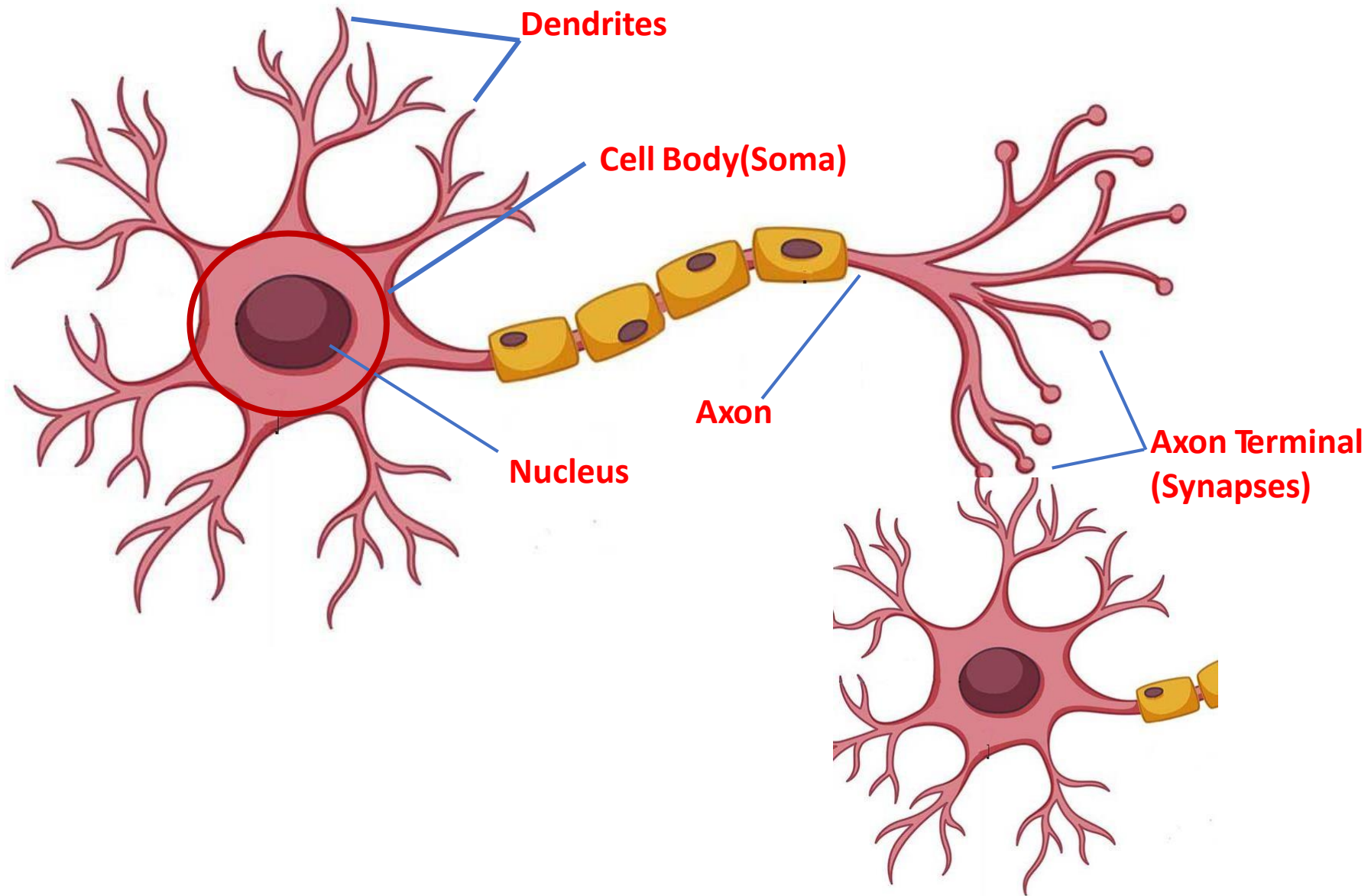
Slides are taken from University
of Pennsylvania

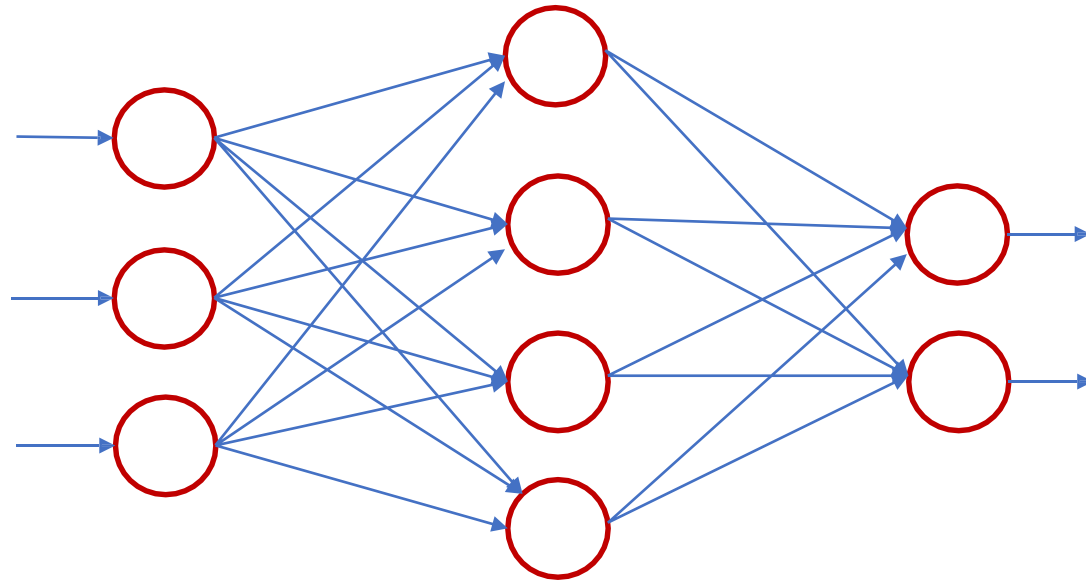
Neural Network: An inspiration from Human Brain

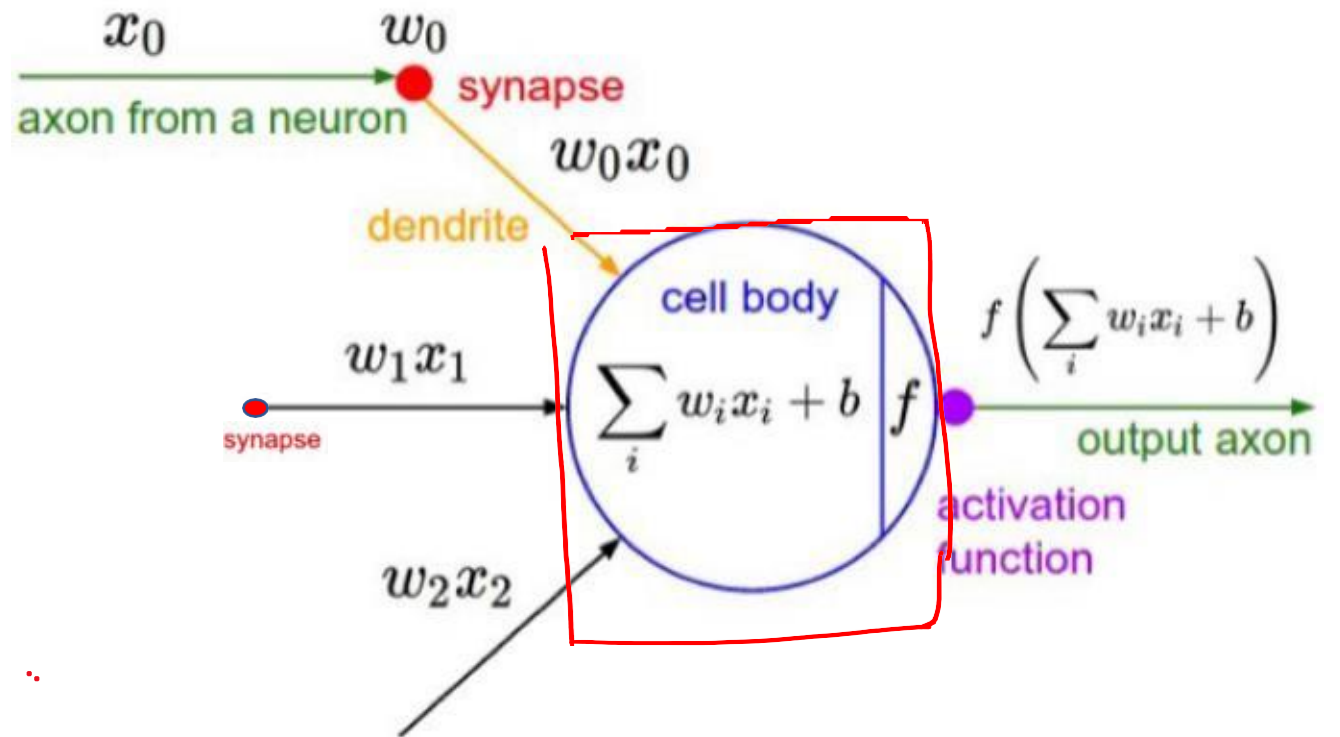
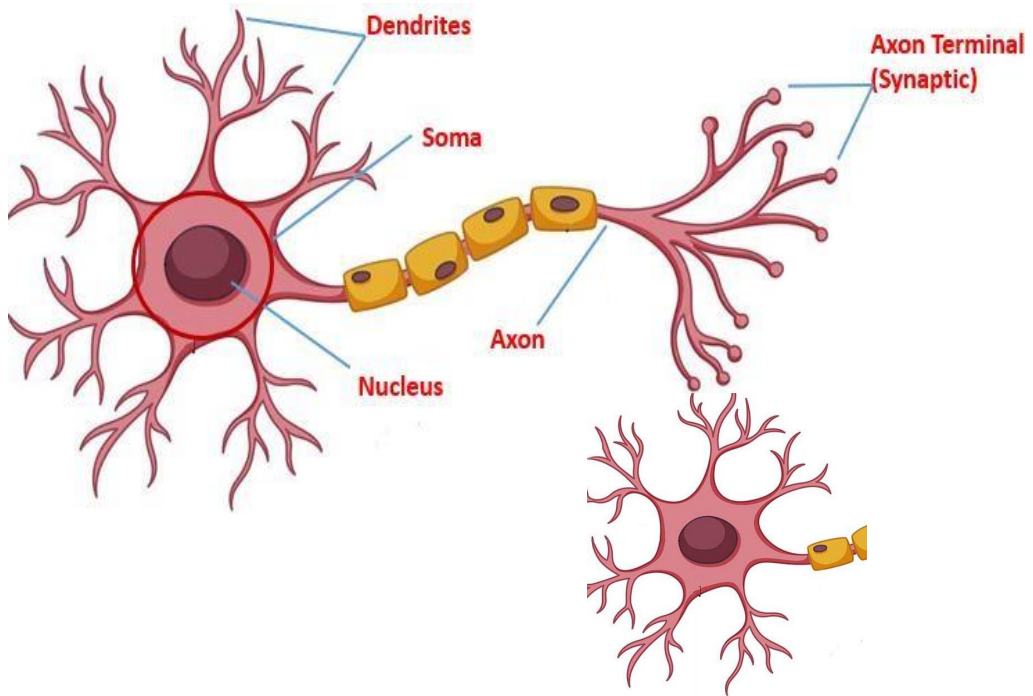


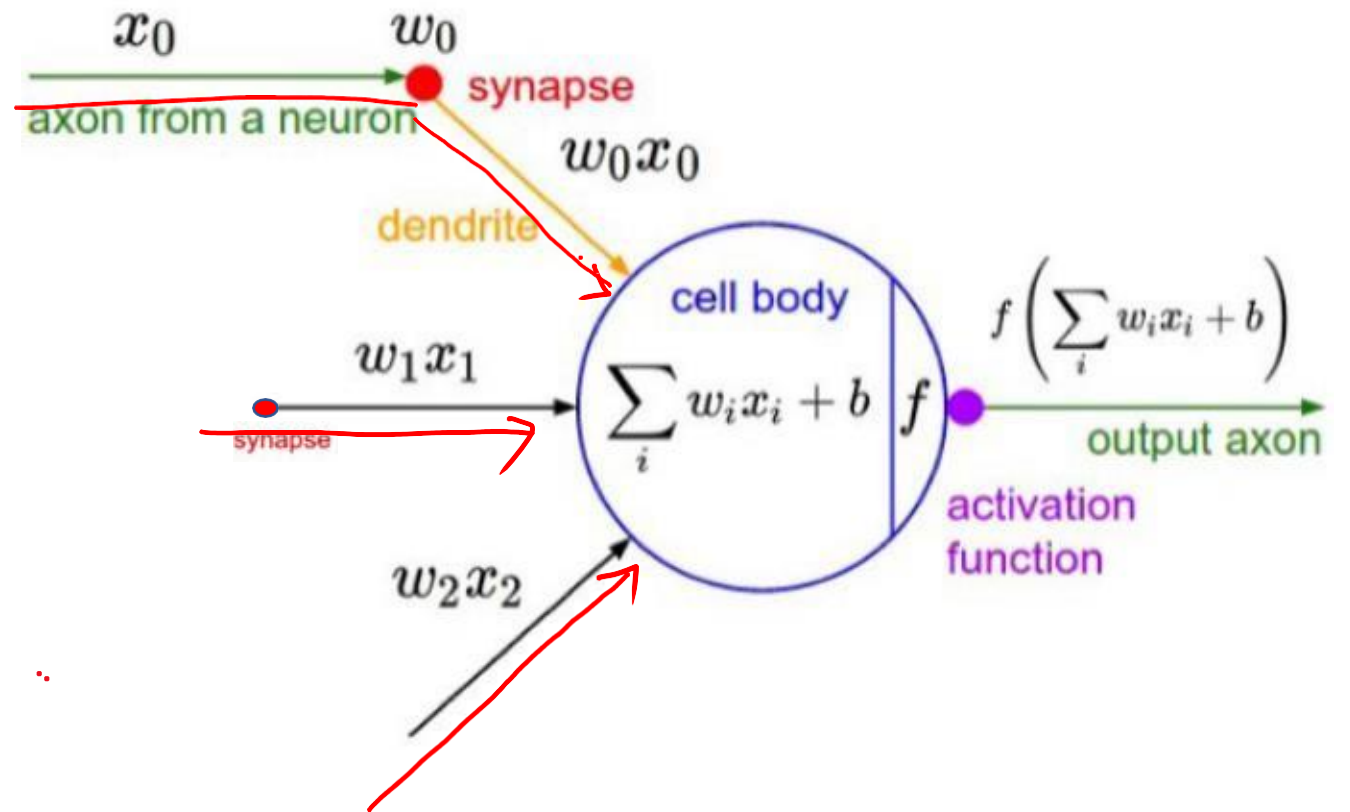
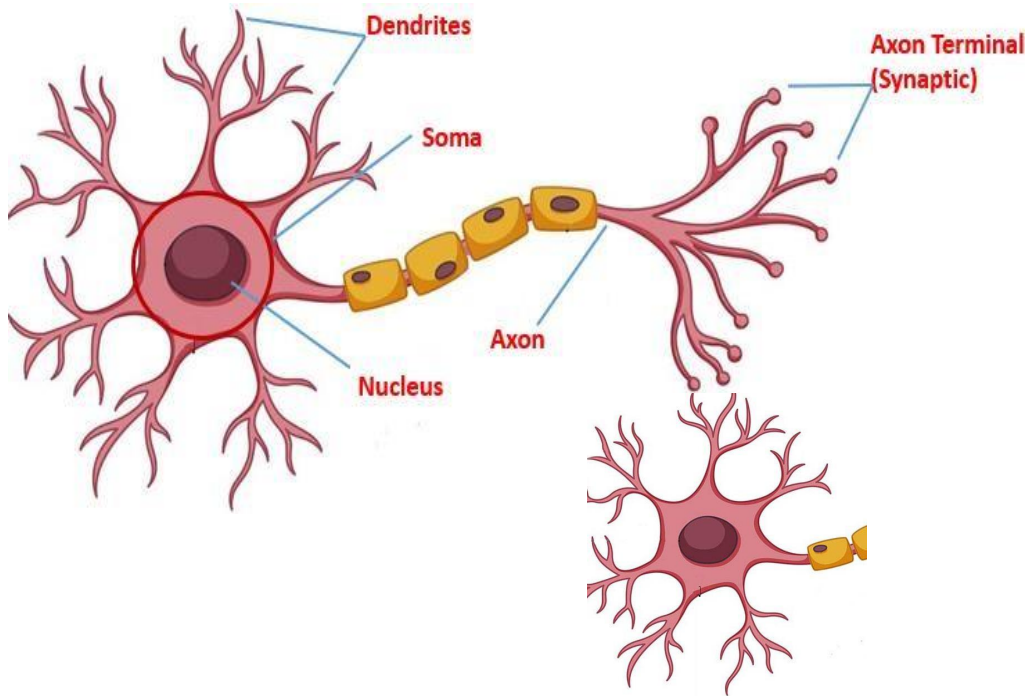


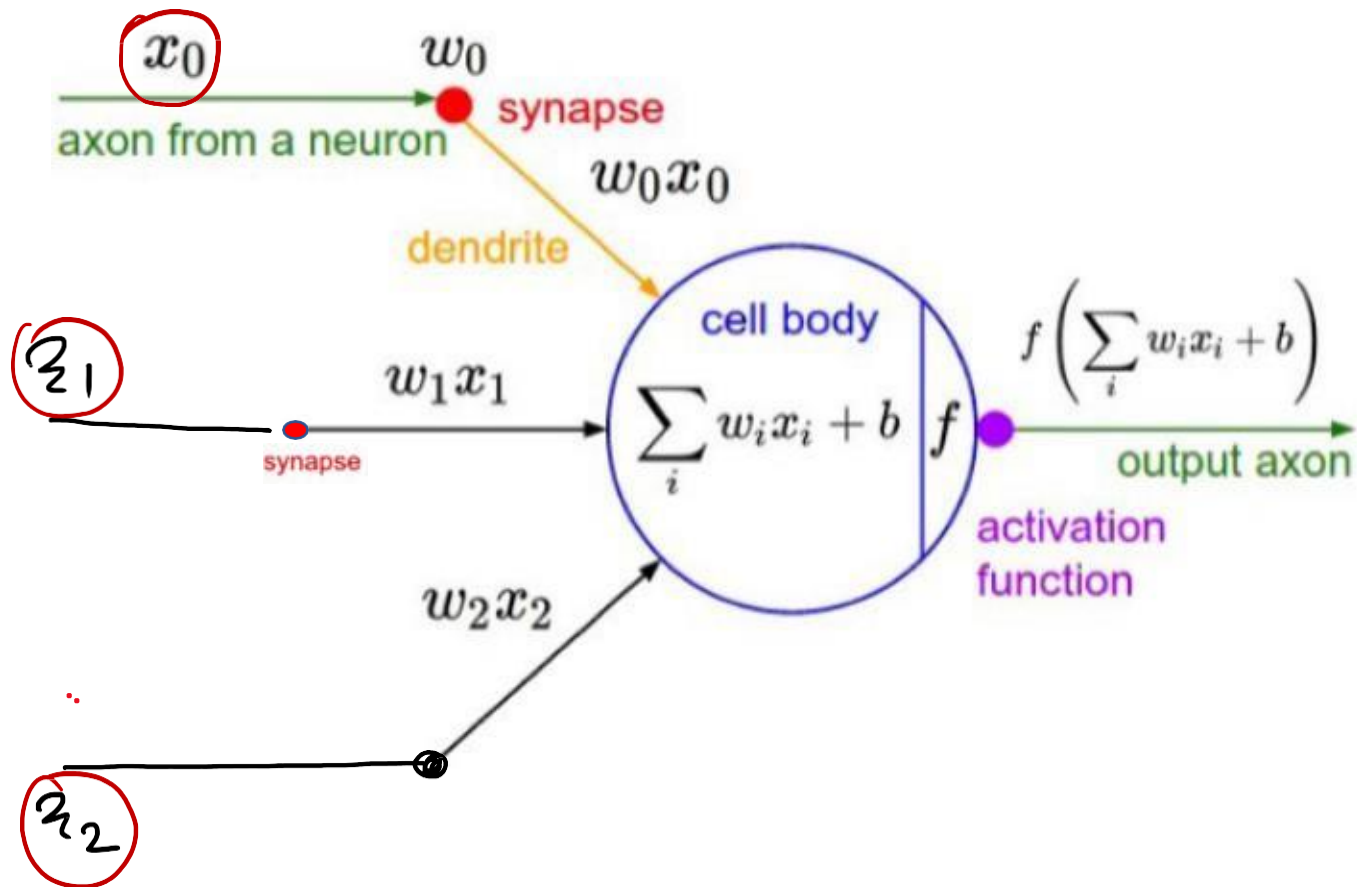
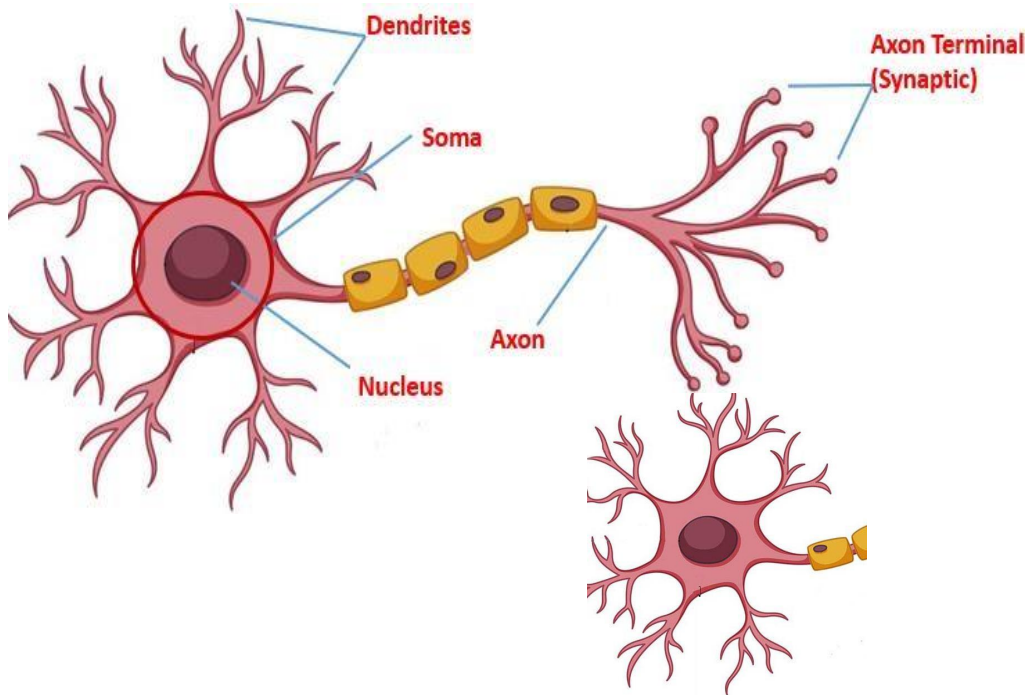
Network of neurons in Human Brain

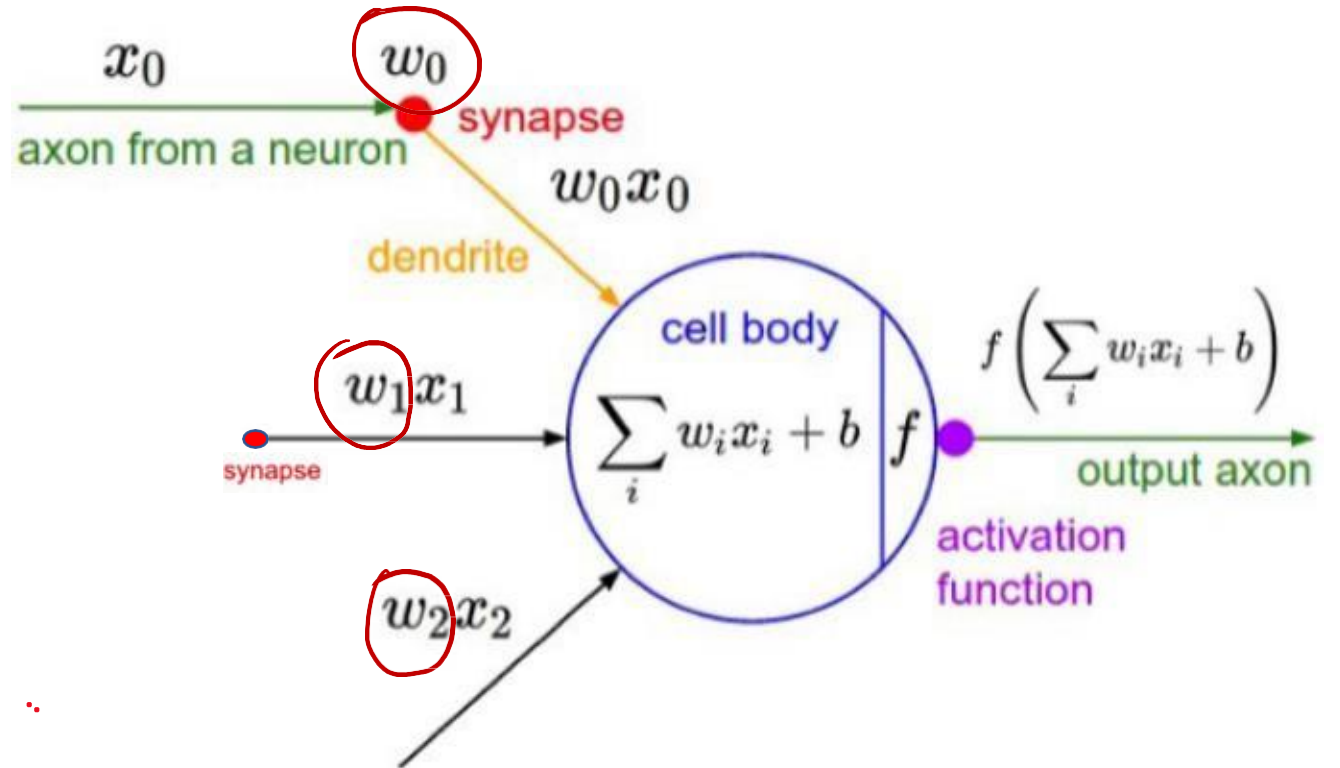
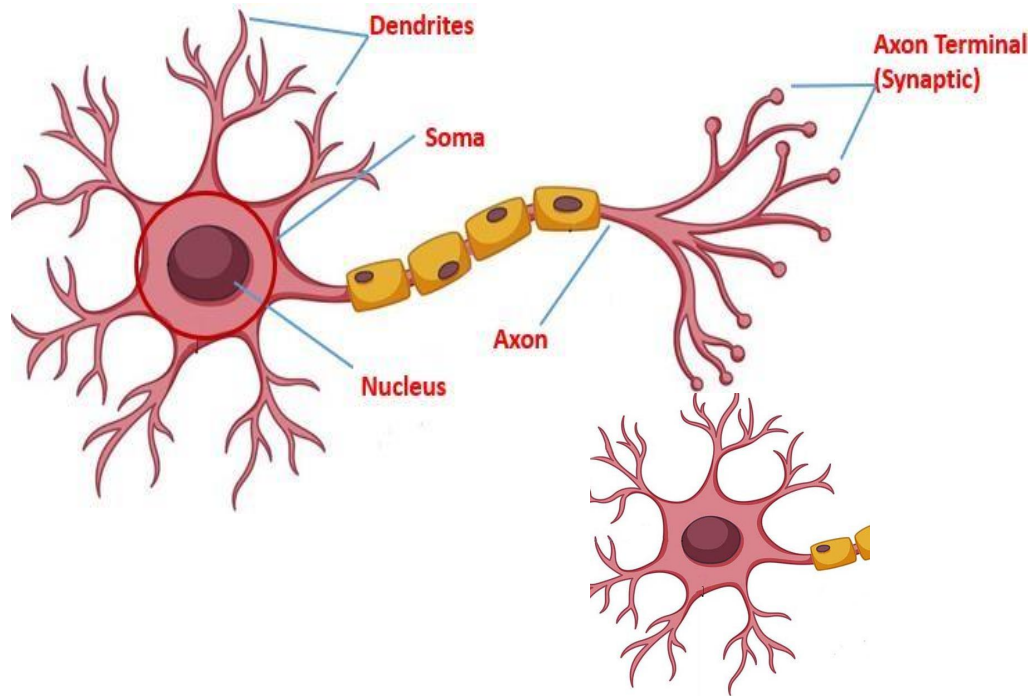


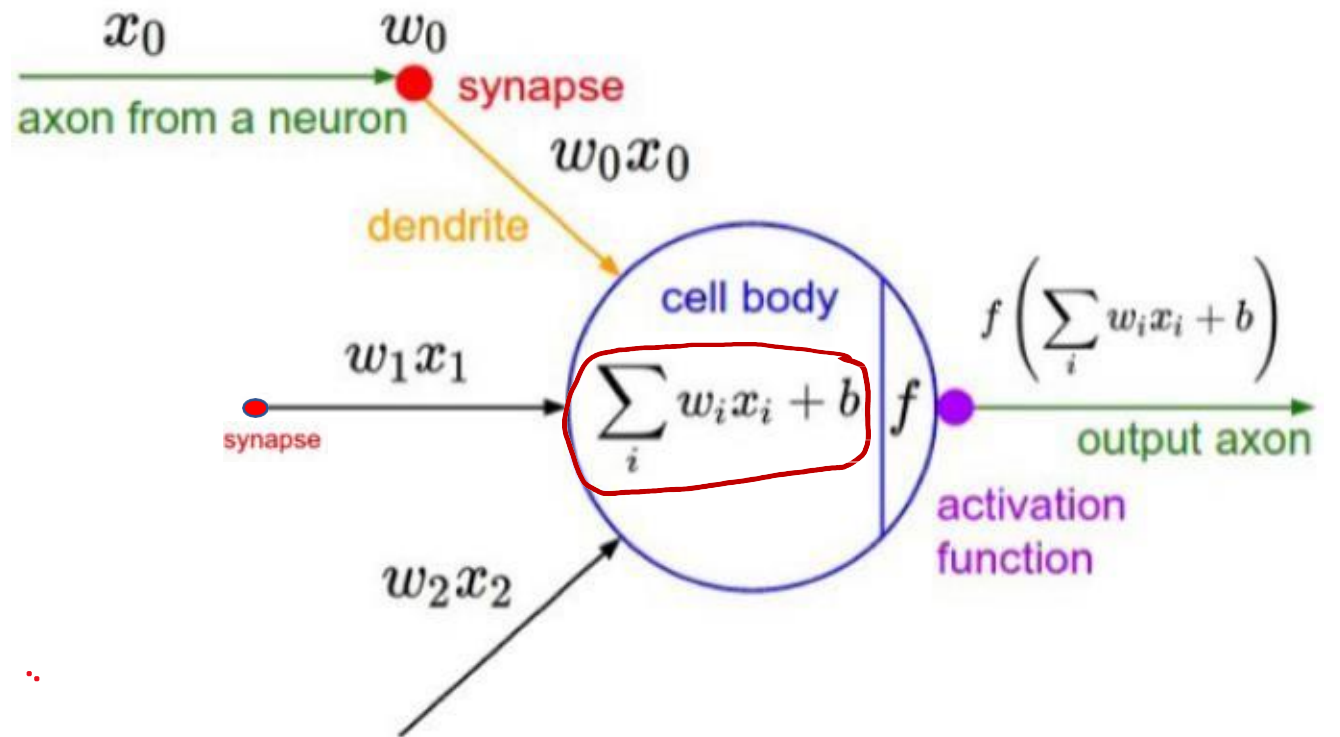
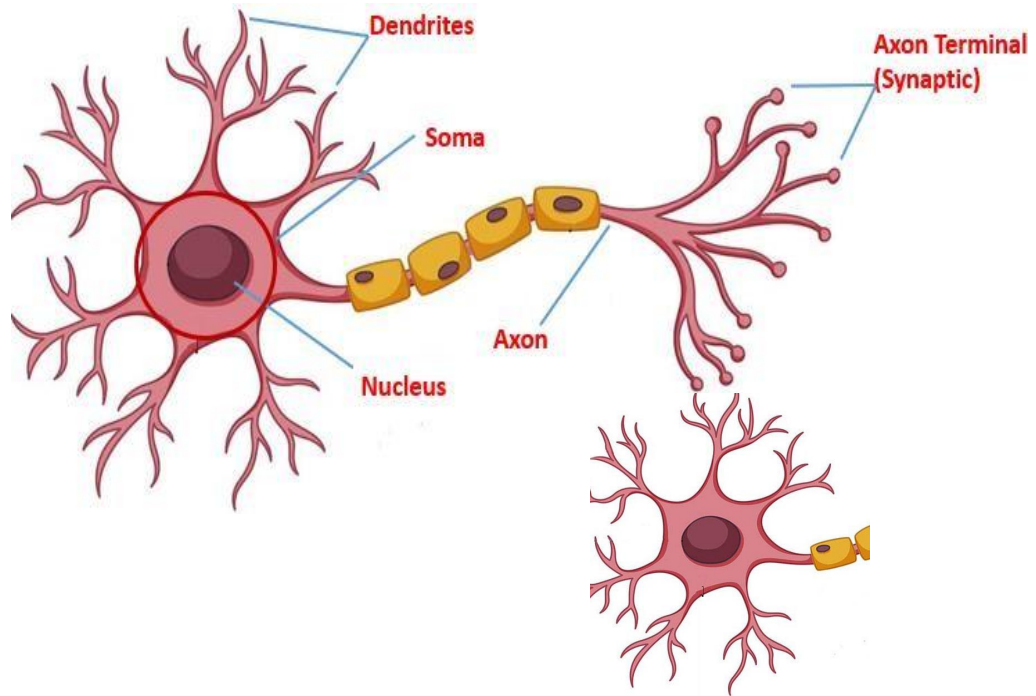


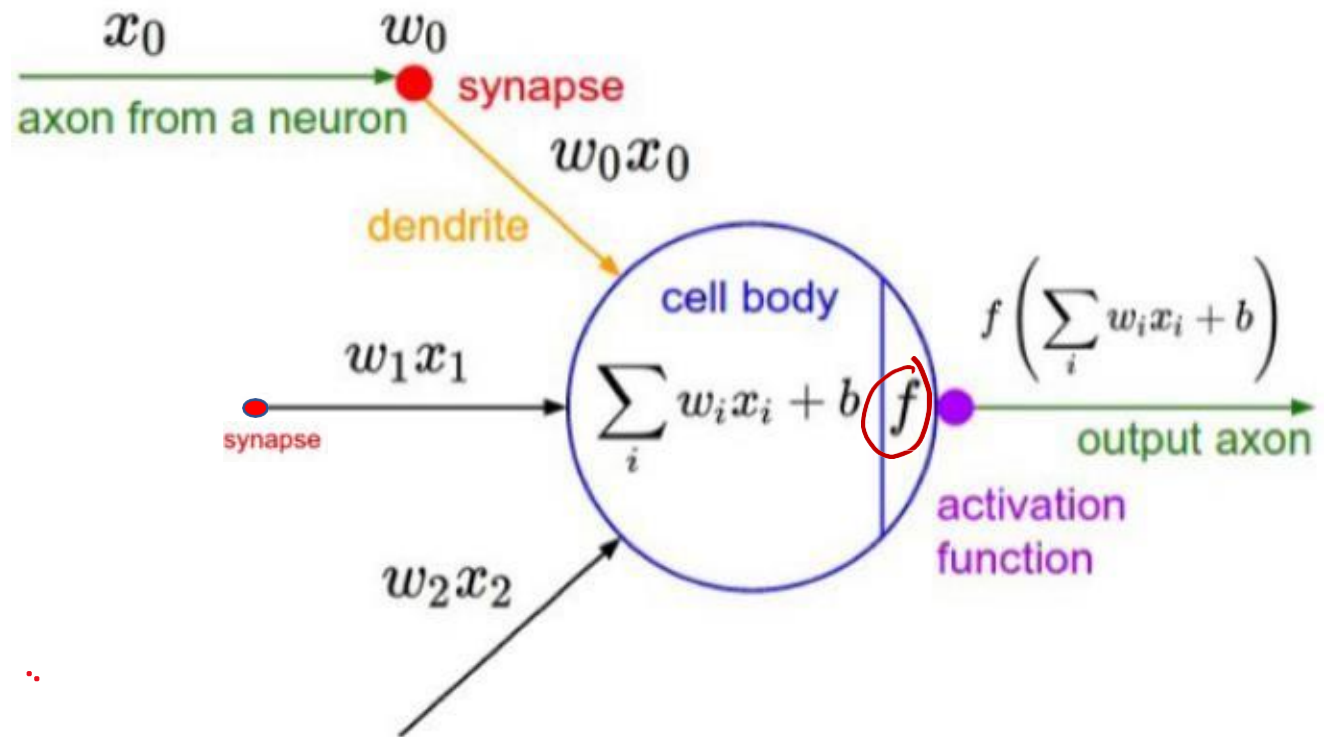
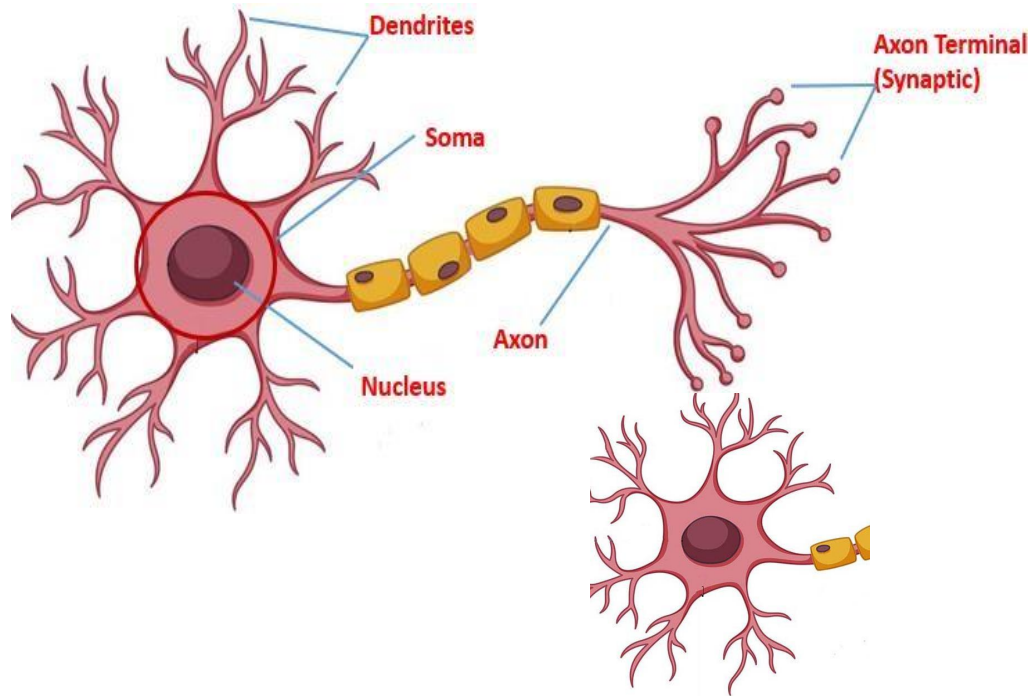


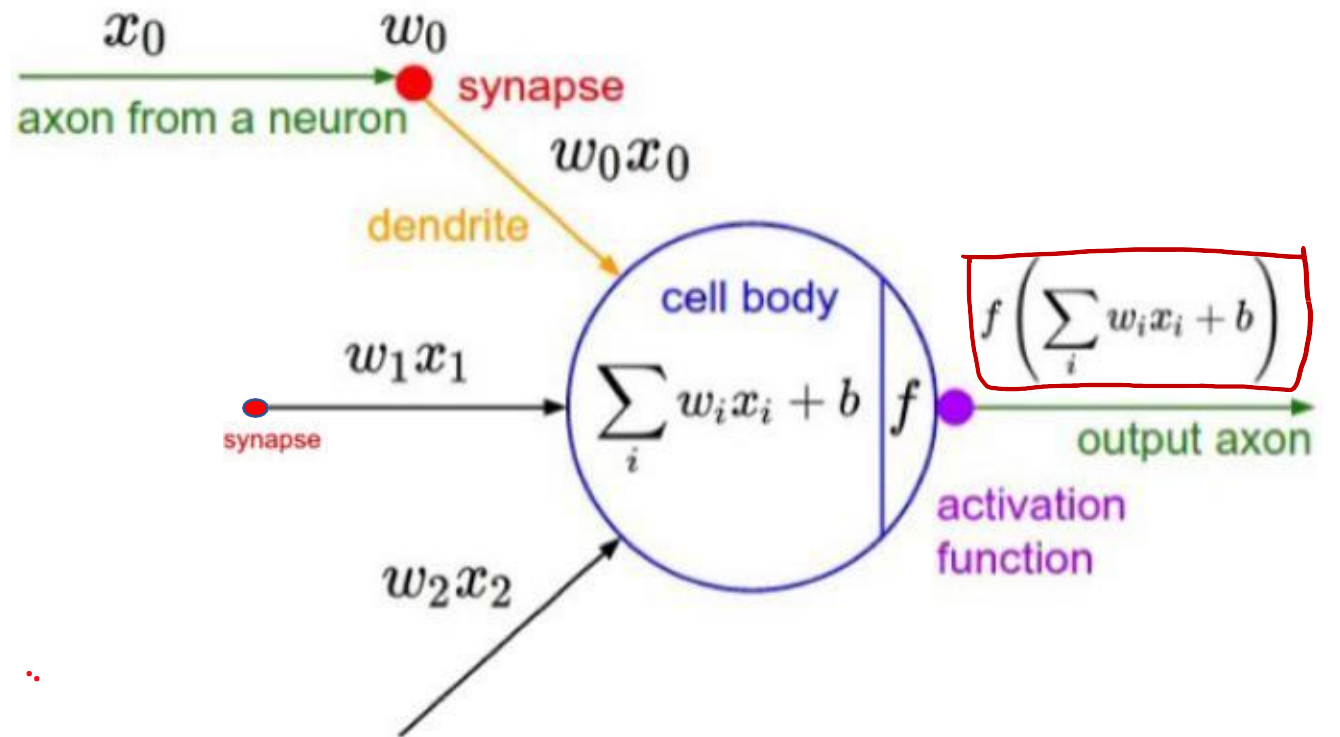
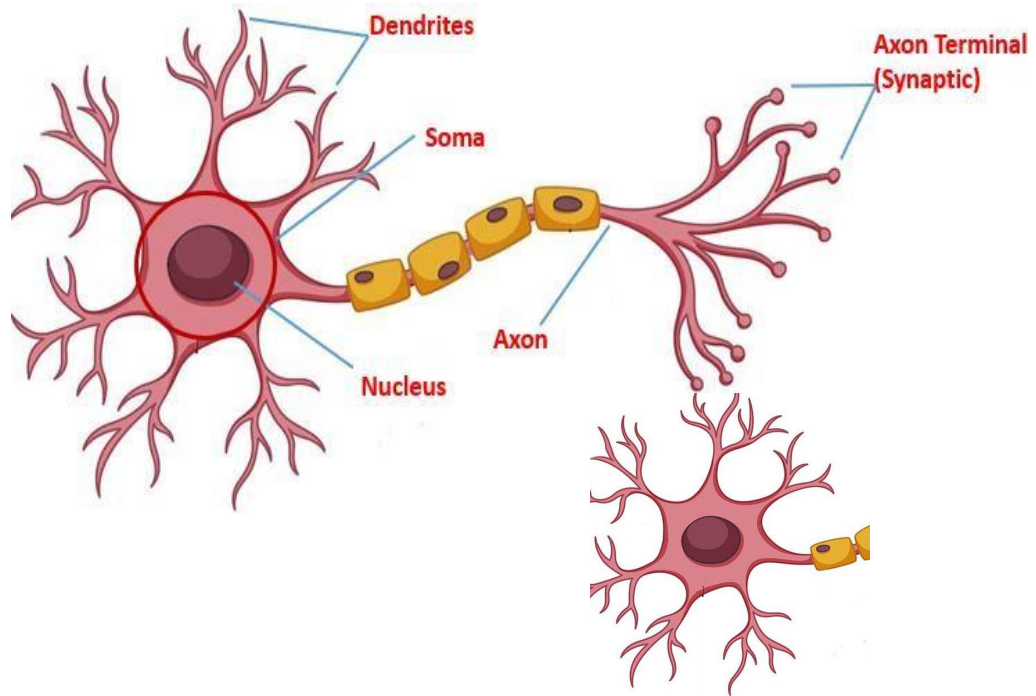






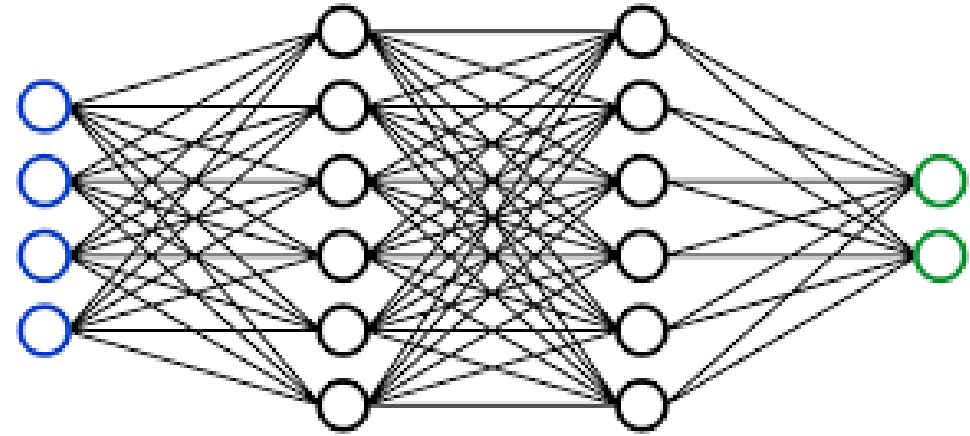
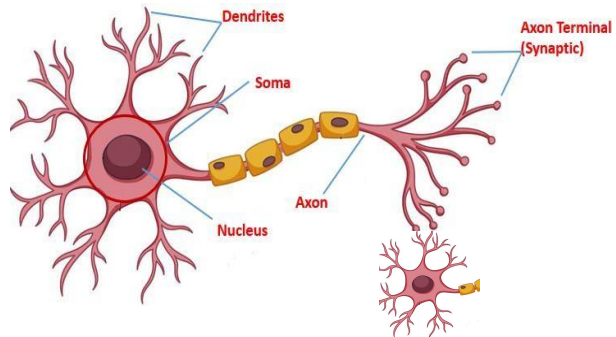






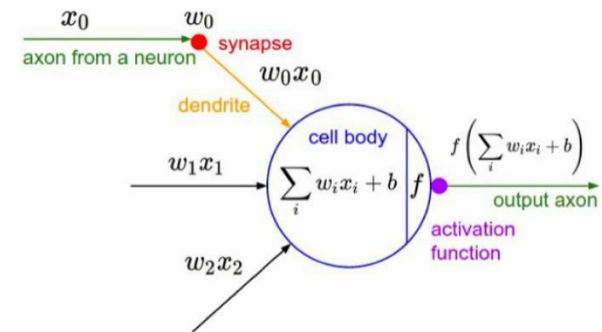


Network of neurons



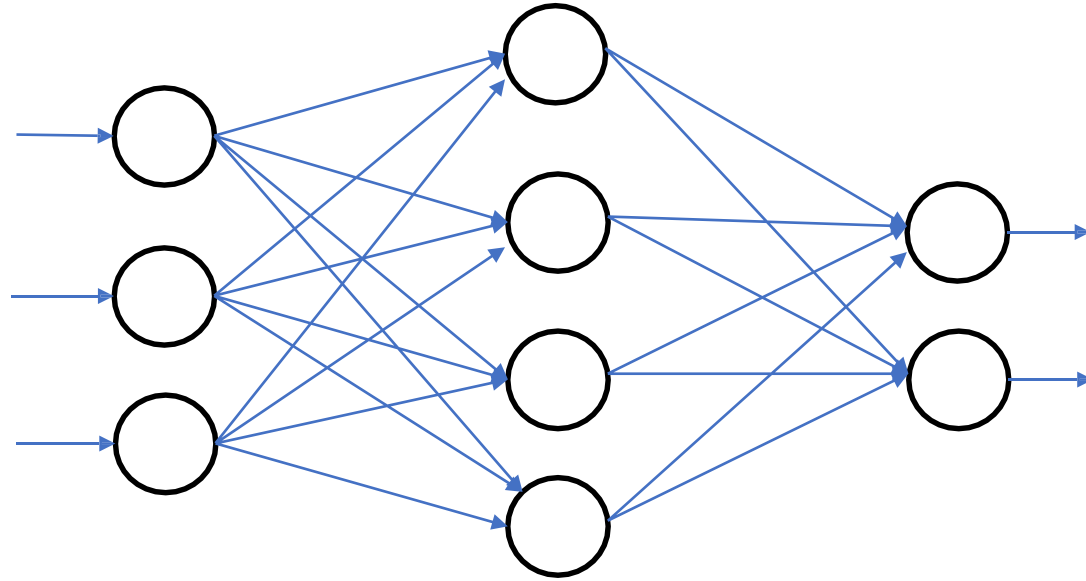
Network of neurons

∴



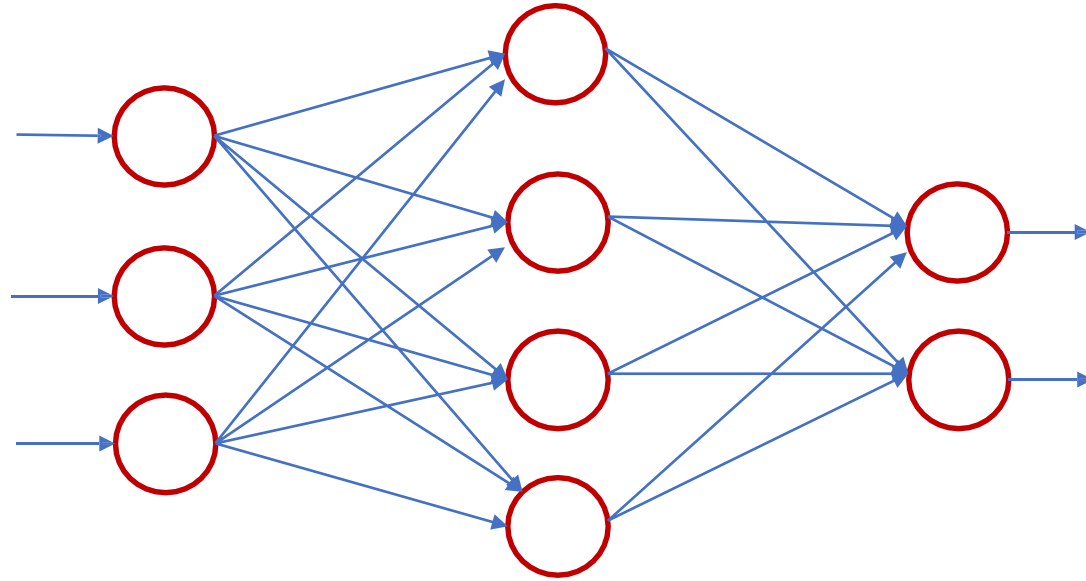
Neural Network: Terminologies and a Toy Example

Terminologies in Artificial Neural Network

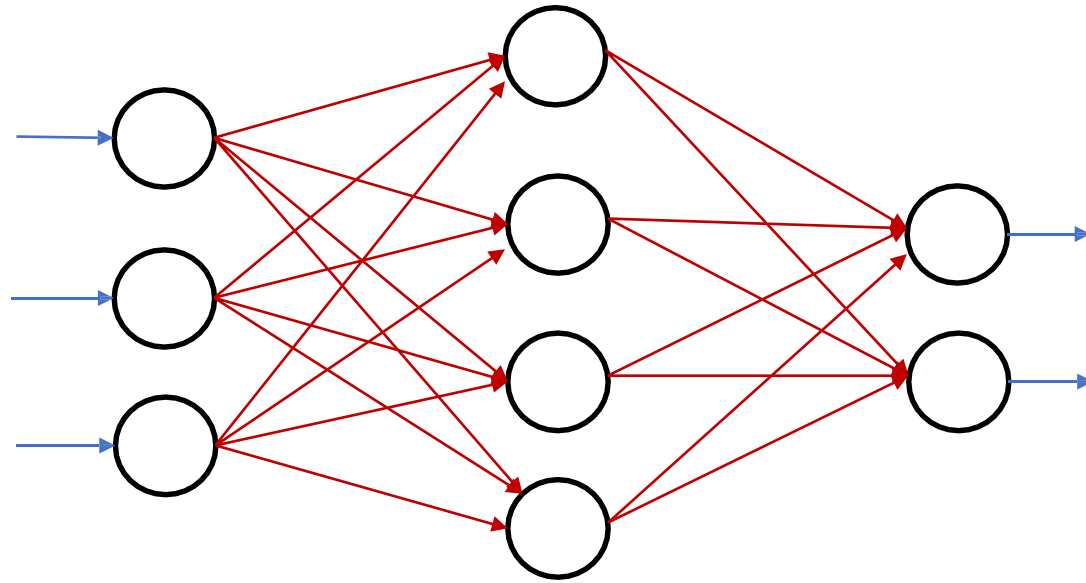


Terminologies in Artificial Neural Network

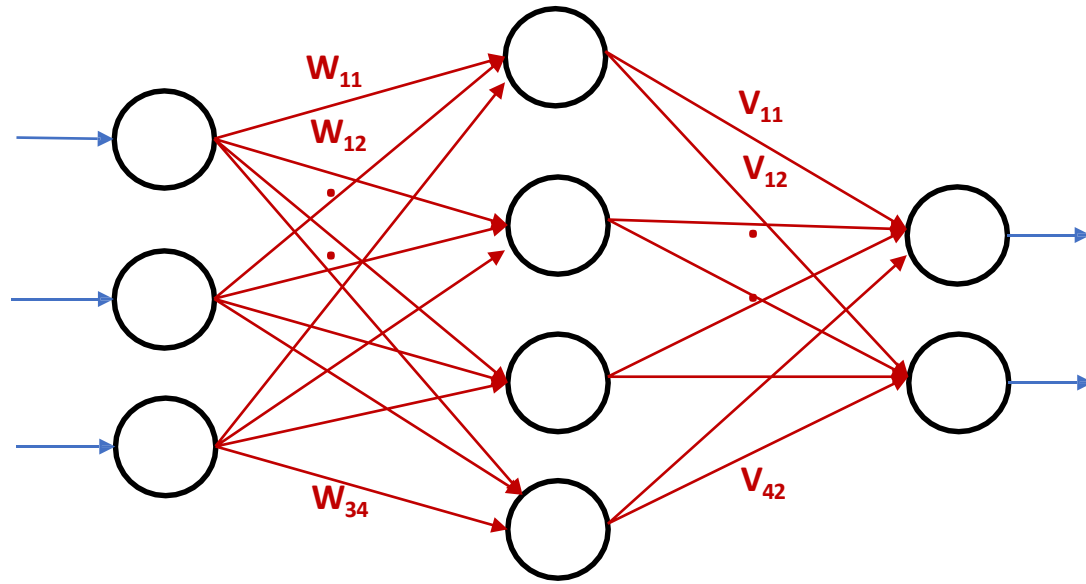
Nodes/Neurons

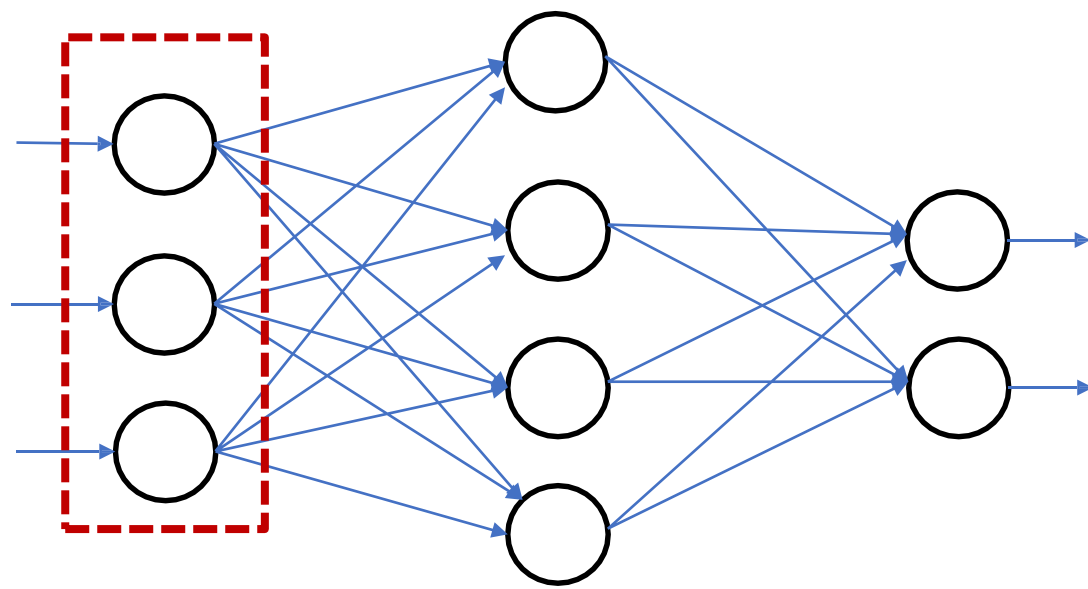


Connections/Dendrites

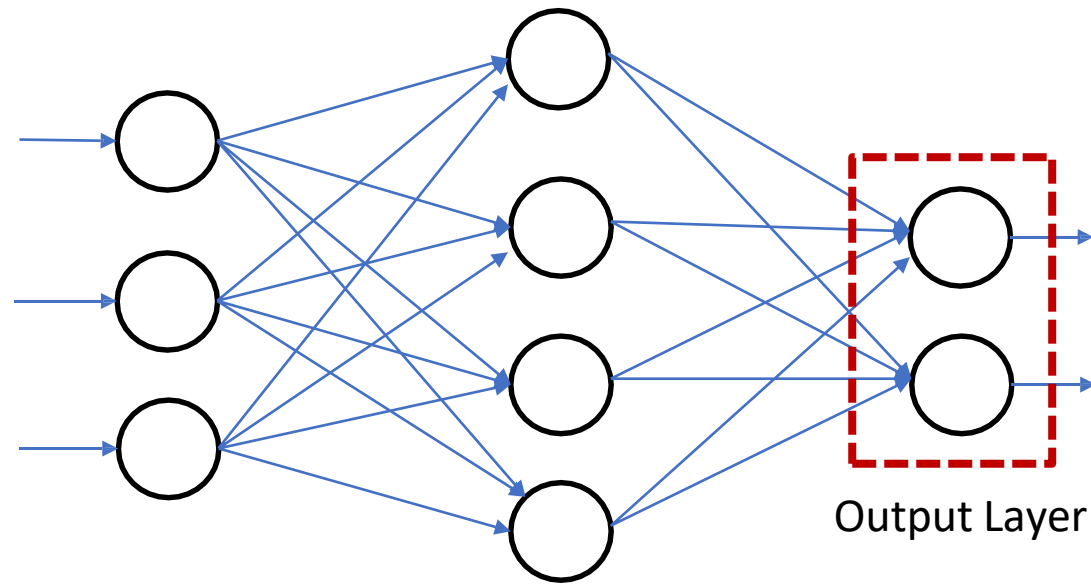


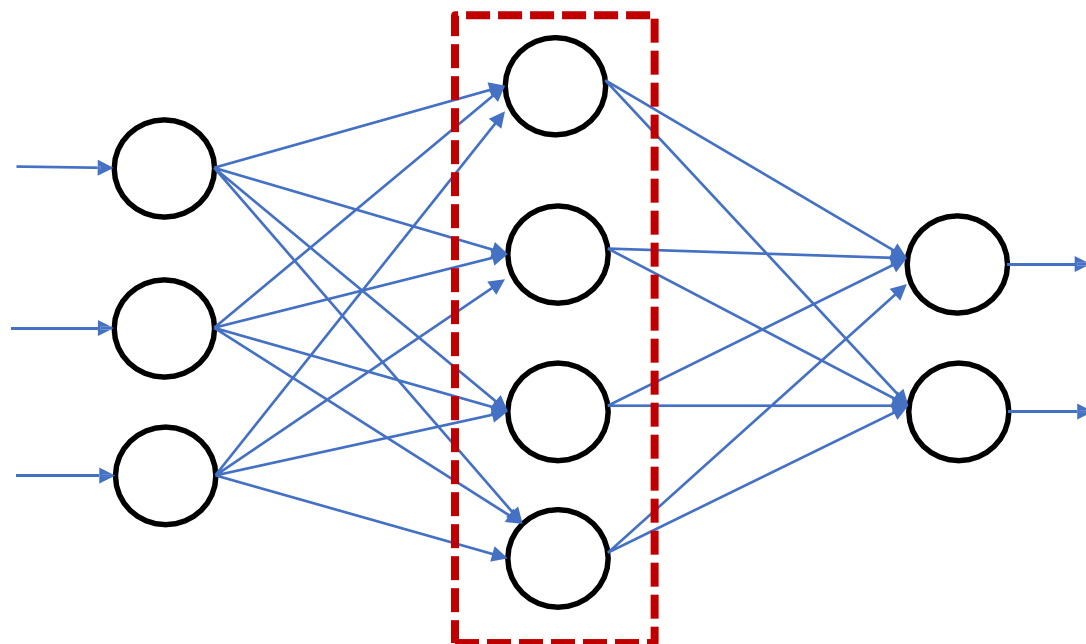
Weights/Synapses



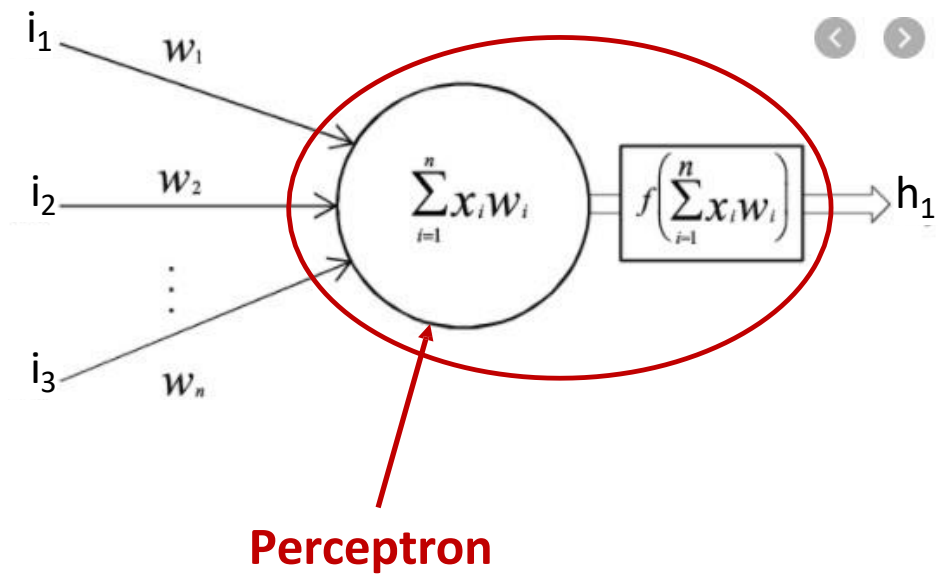
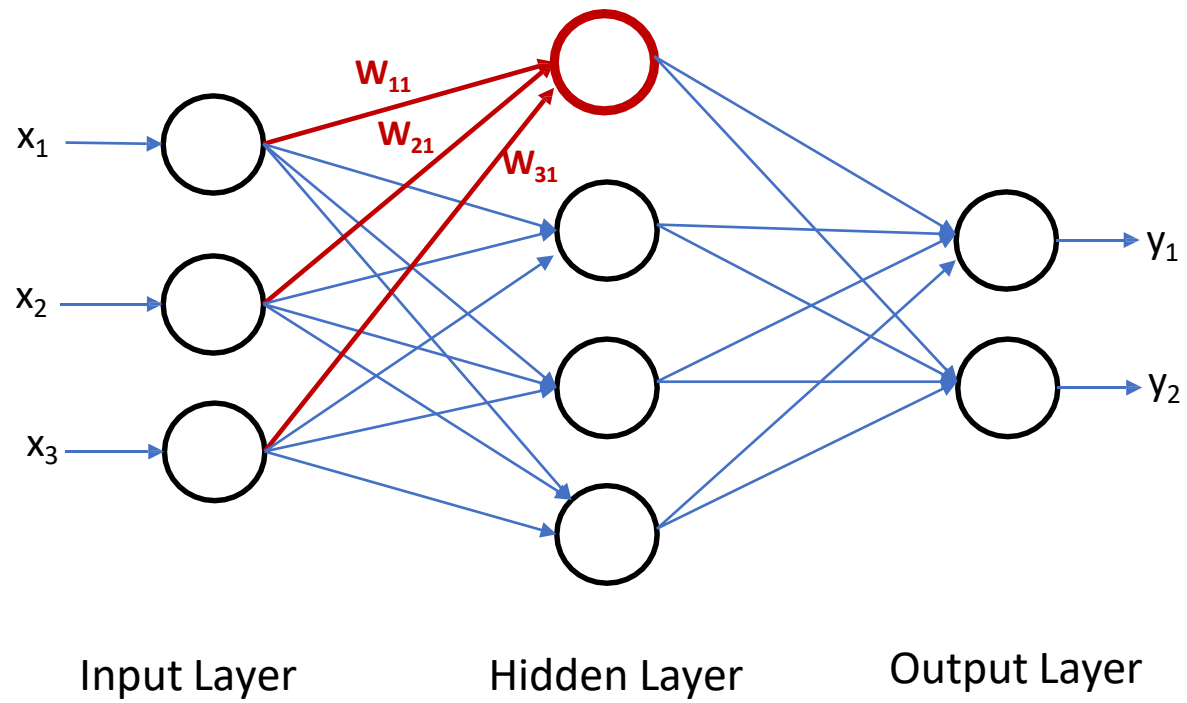


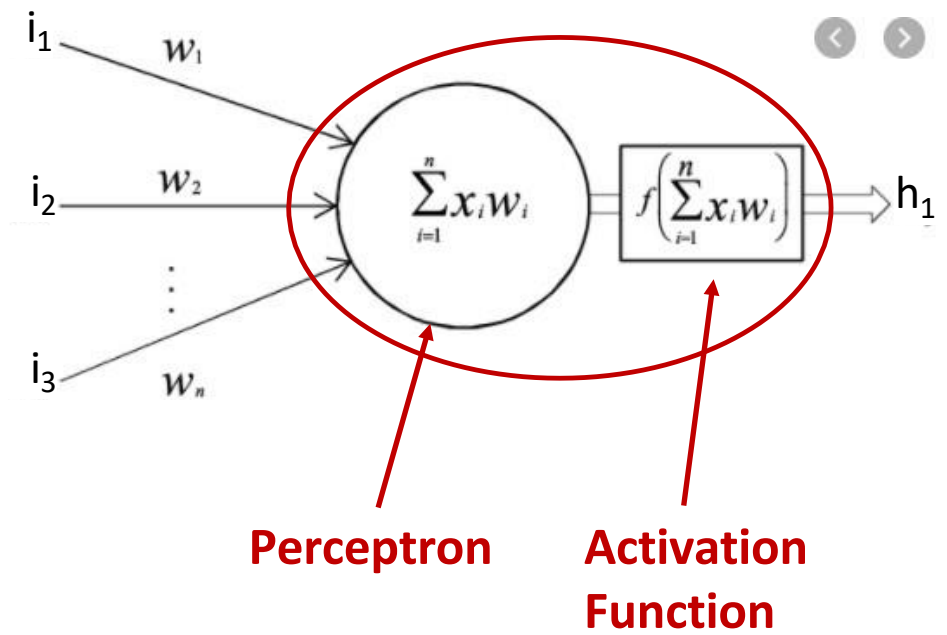
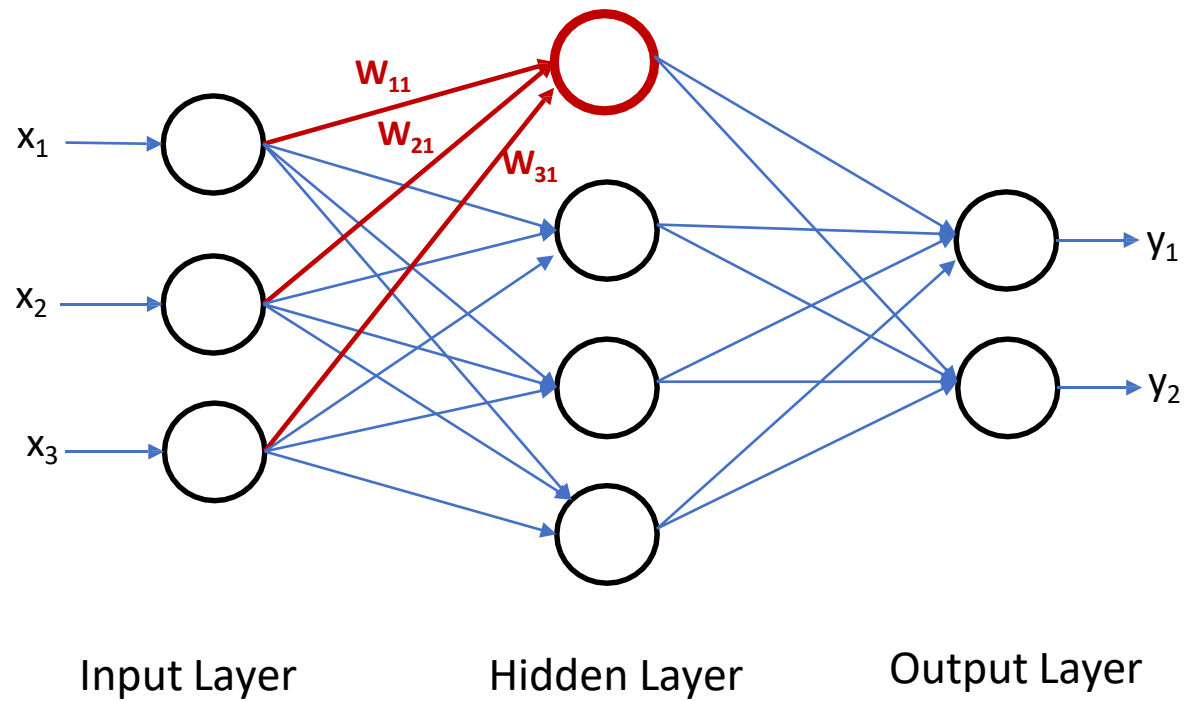
Input Layer



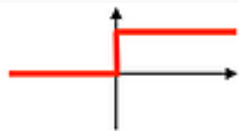
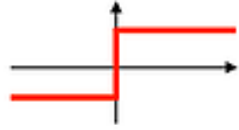
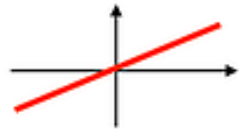
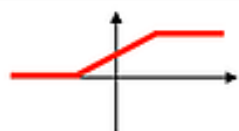
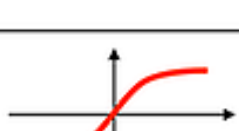


Hidden Layer





Activation Functions

Activation function	Equation	Example	1D Graph
Unit step (Heaviside)	$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Sign (Signum)	$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$	Perceptron variant	
Linear	$\phi(z) = z$	Adaline, linear regression	
Piece-wise linear	$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multi-layer NN	
Hyperbolic tangent	$\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	Multi-layer Neural Networks	
Rectifier, ReLU (Rectified Linear Unit)	$\phi(z) = \max(0, z)$	Multi-layer Neural Networks	
Rectifier, softplus	$\phi(z) = \ln(1 + e^z)$	Multi-layer Neural Networks	

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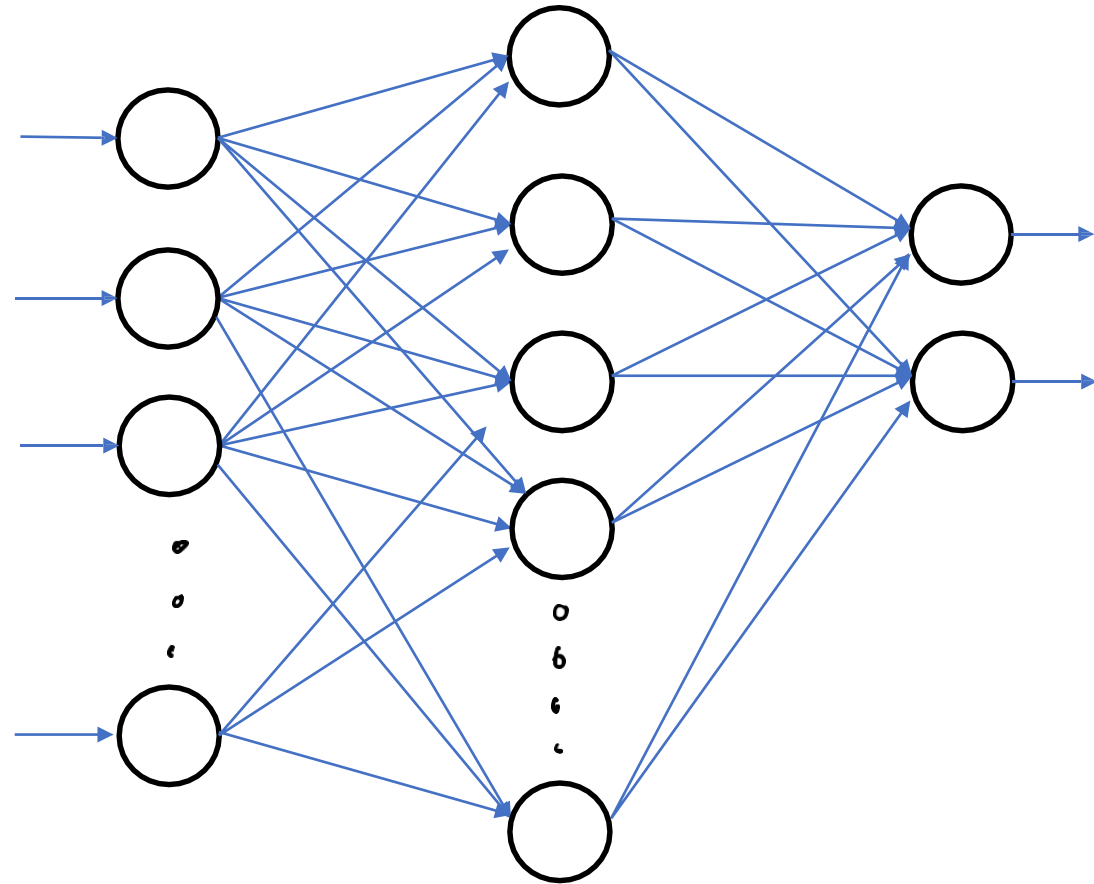
A Toy Example



60x60=3600



60x60=3600



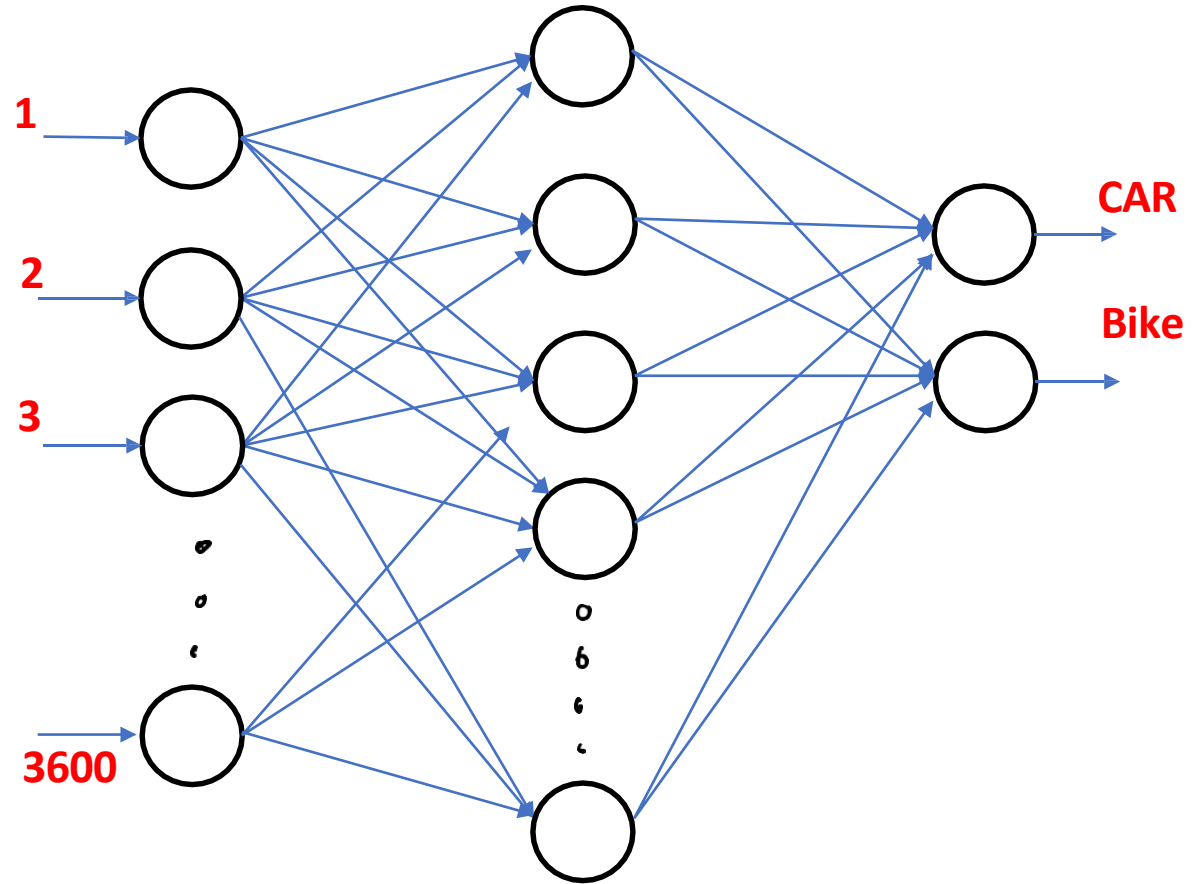
A Toy Example



60x60=3600



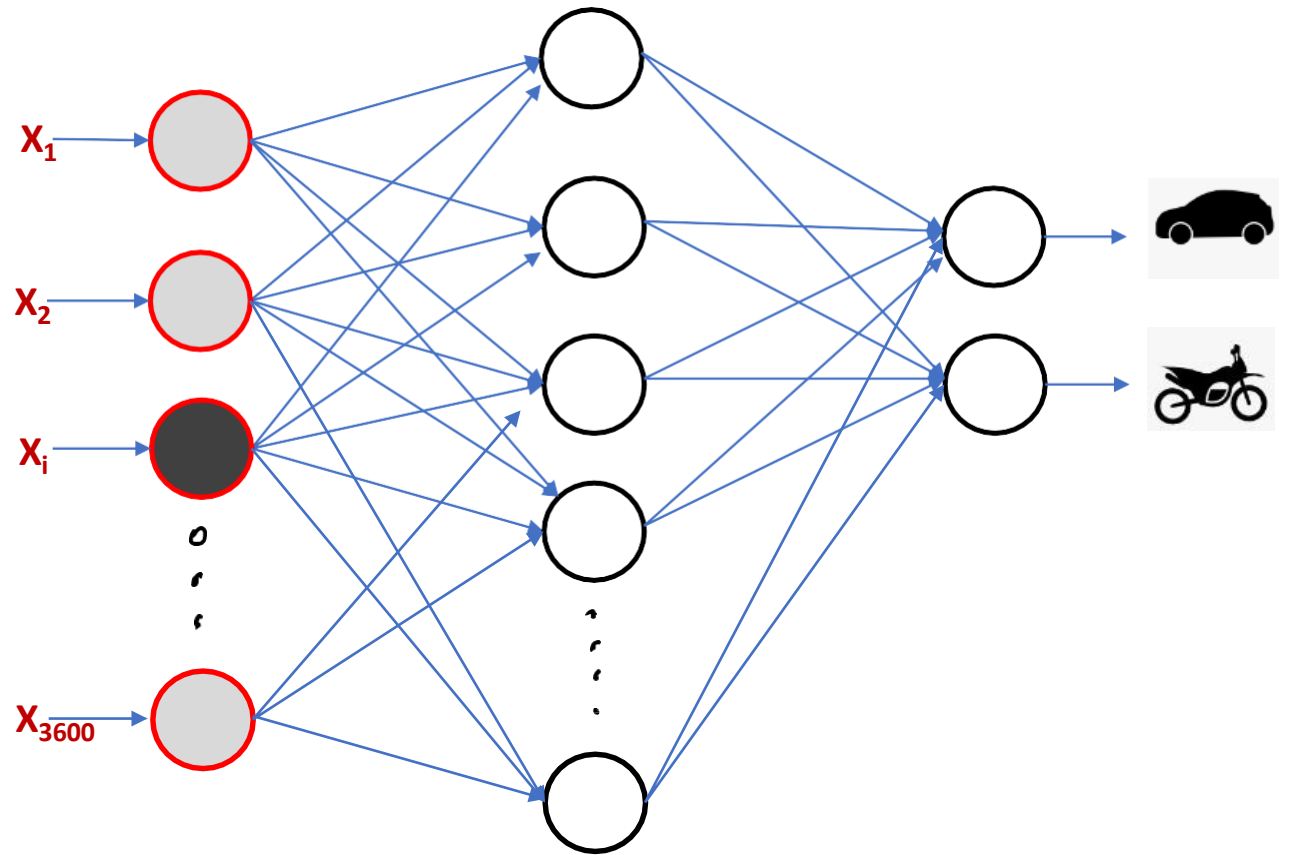
60x60=3600



A Toy Example



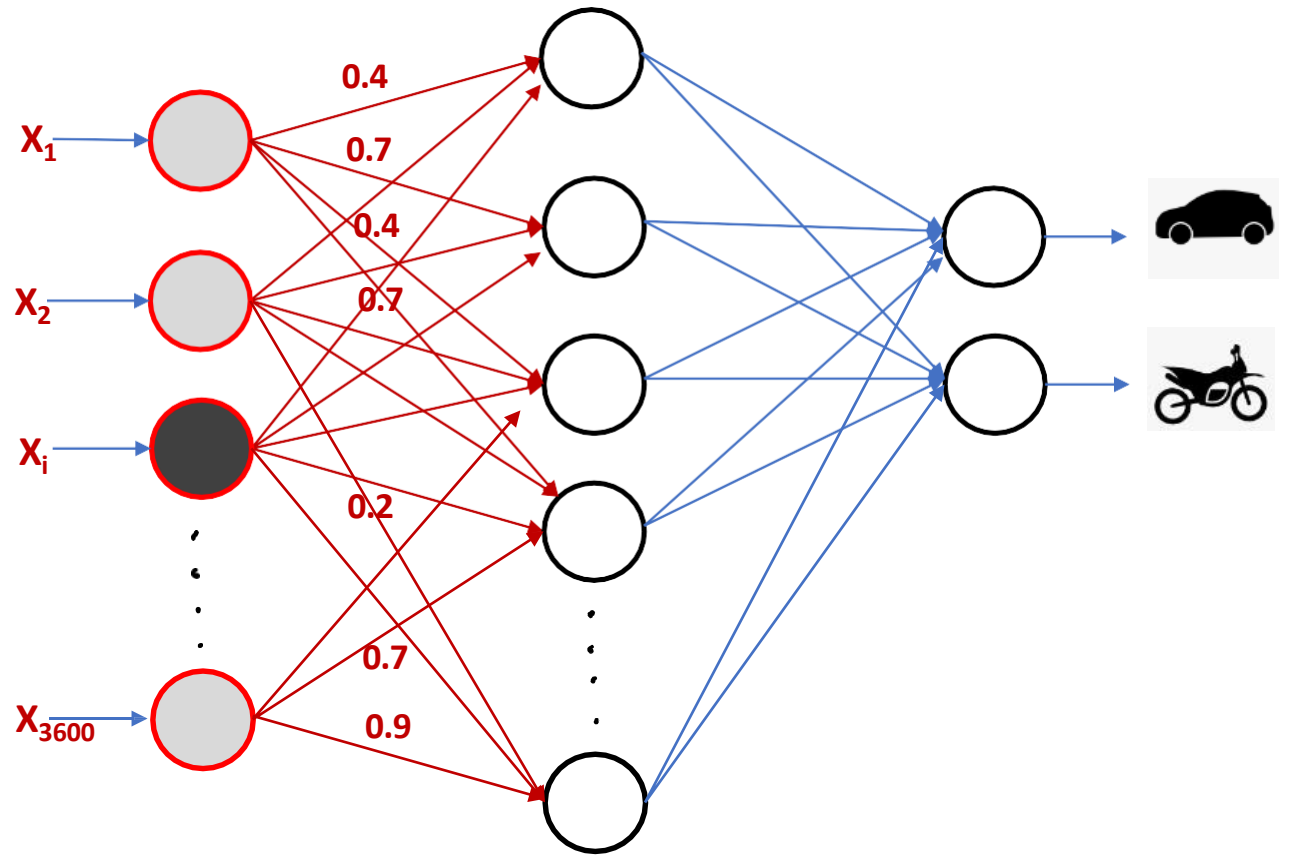
$60 \times 60 = 3600$



A Toy Example



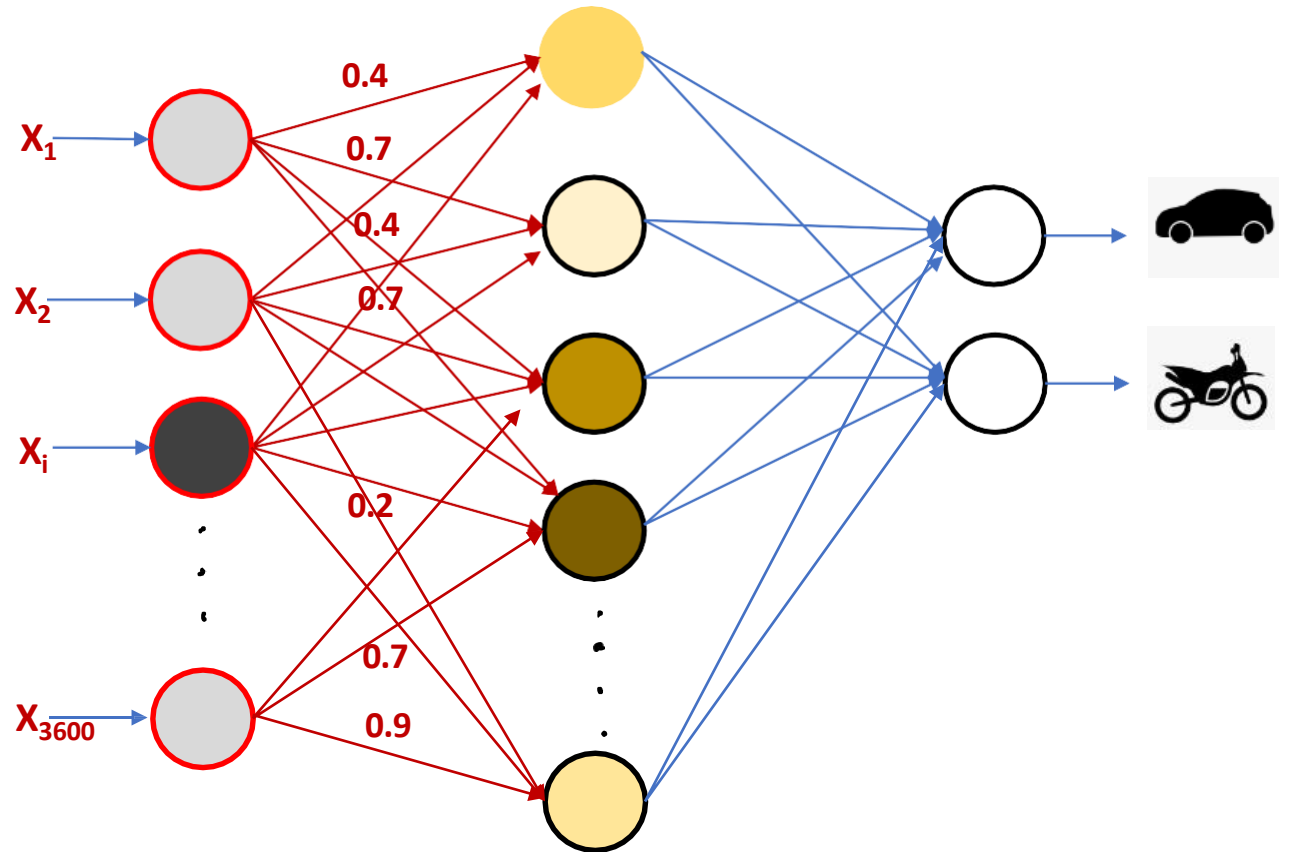
60x60=3600



A Toy Example



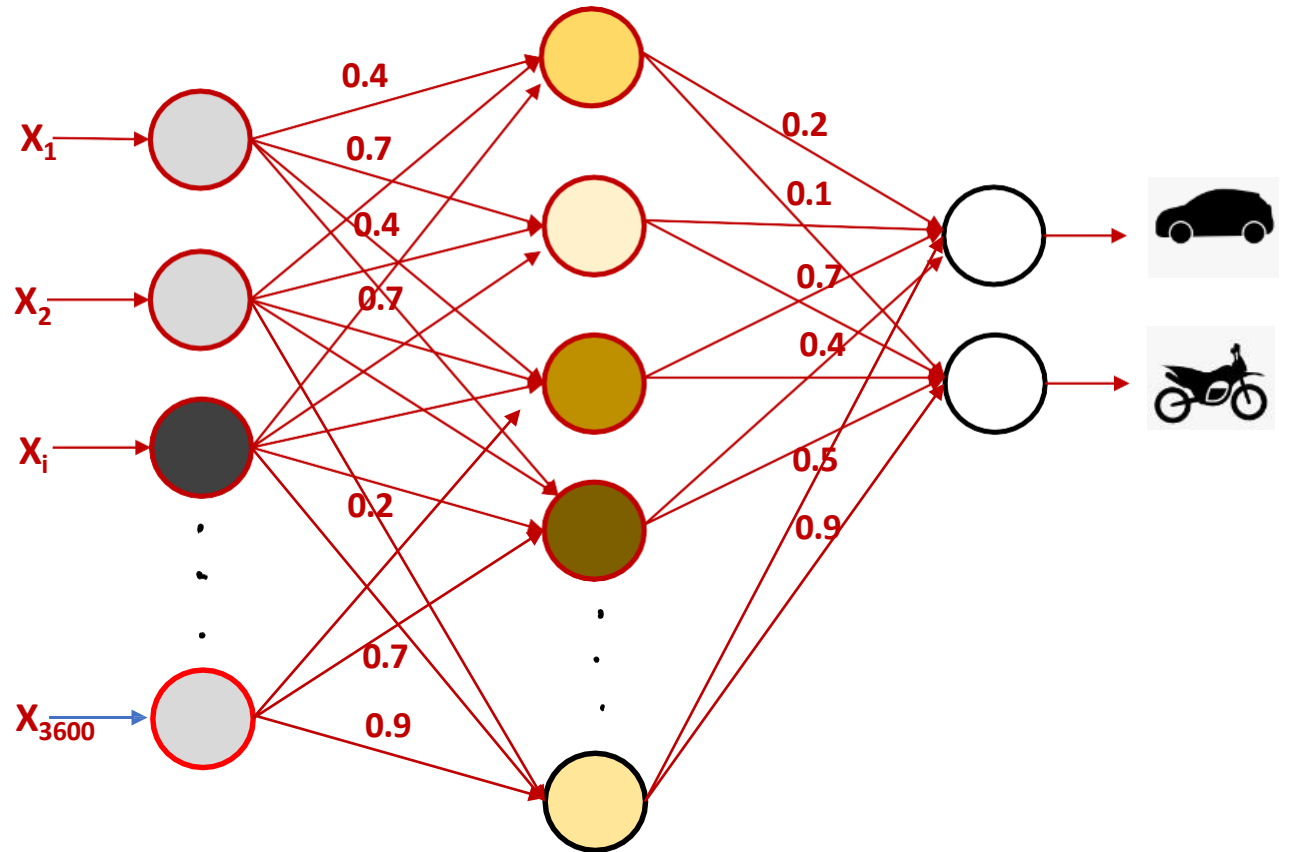
60x60=3600



A Toy Example



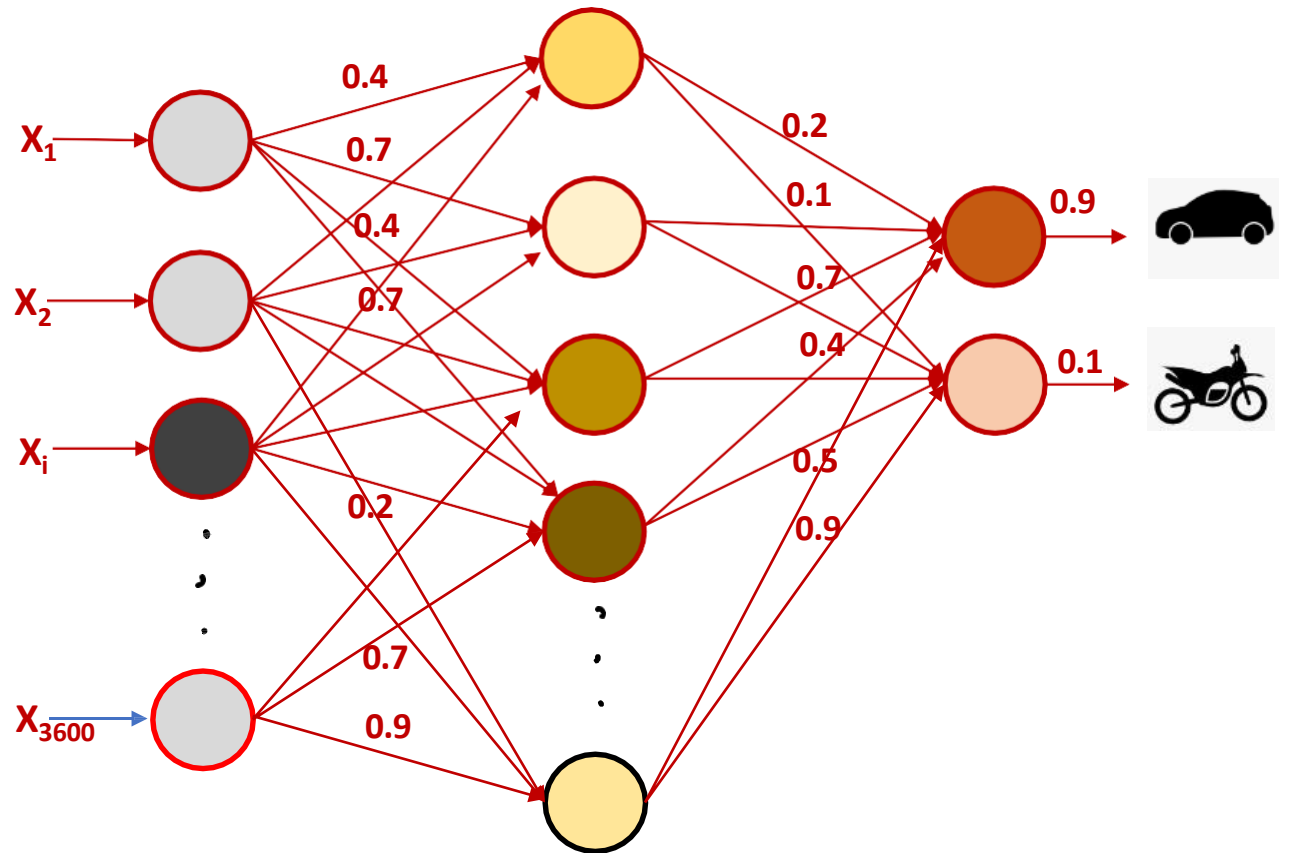
60x60=3600



A Toy Example



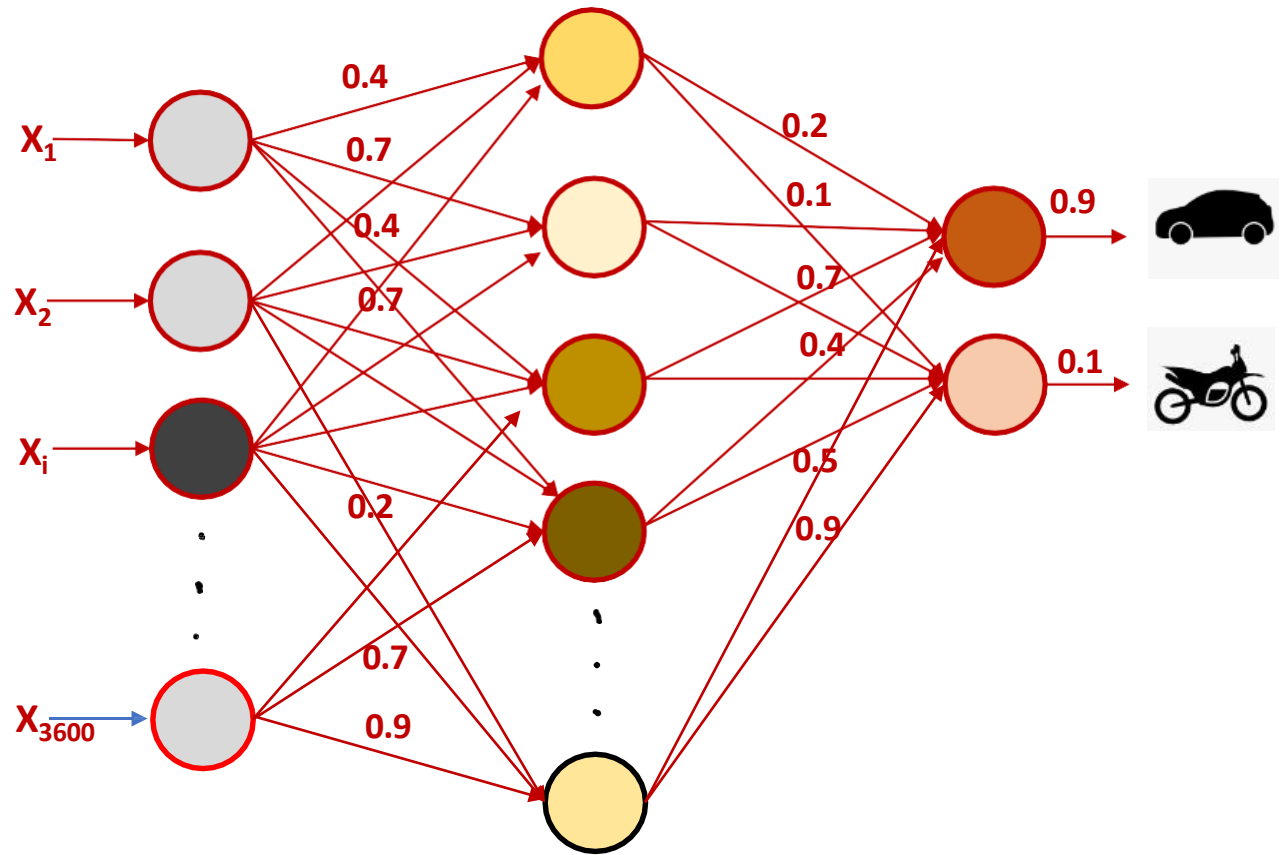
60x60=3600



A Toy Example



60x60=3600

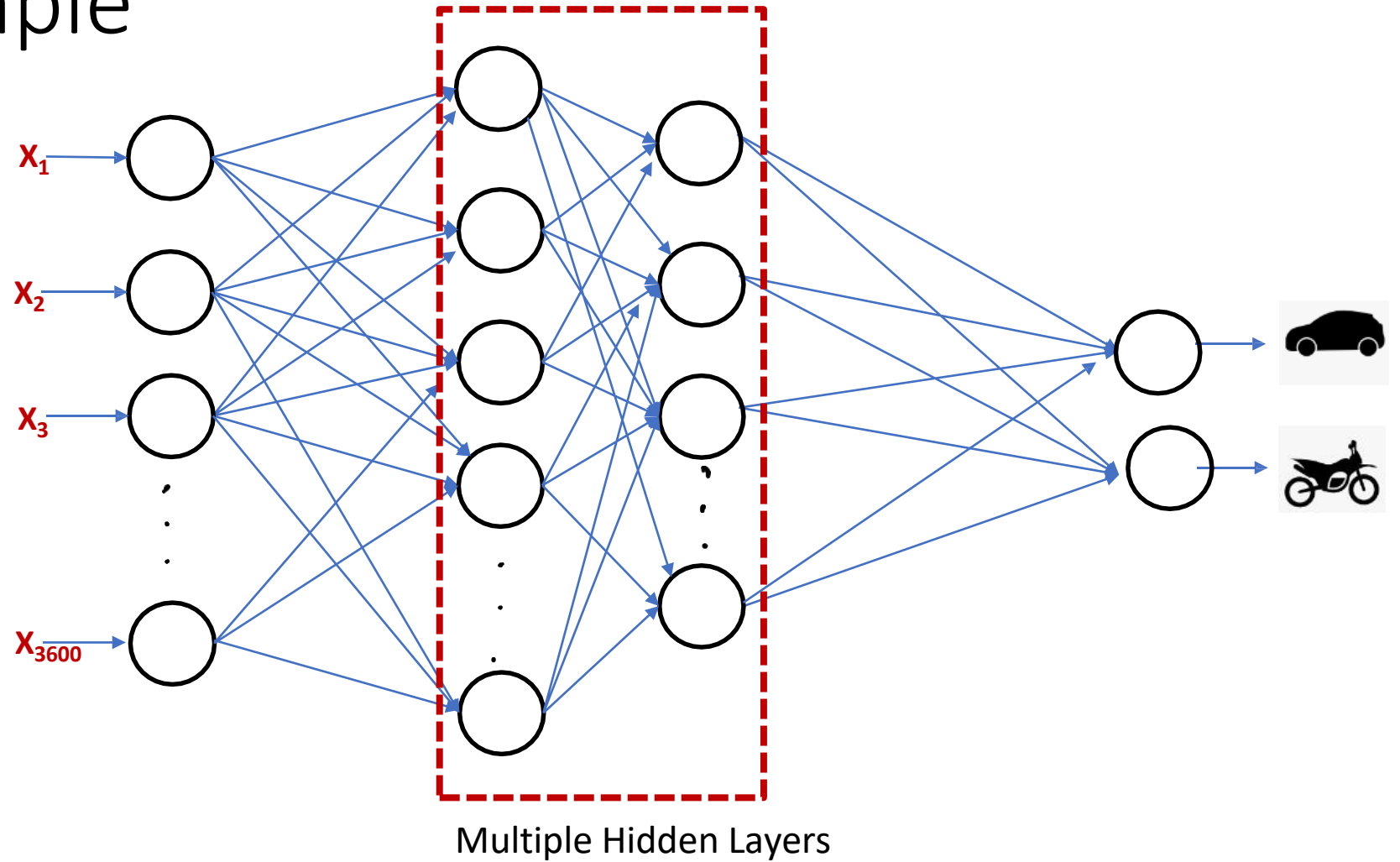


Forward pass

A Toy Example



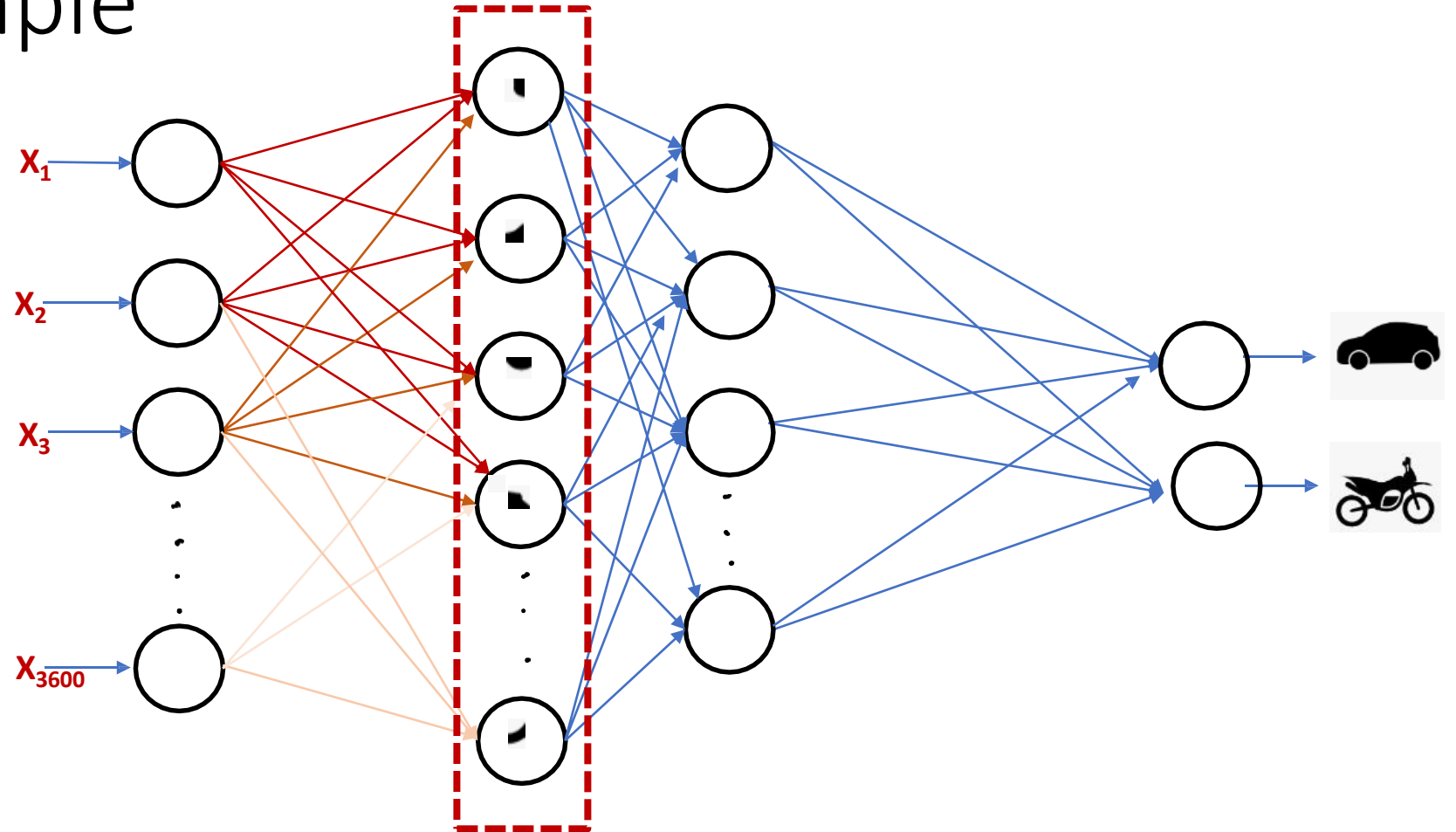
60x60=3600



A Toy Example



60x60=3600

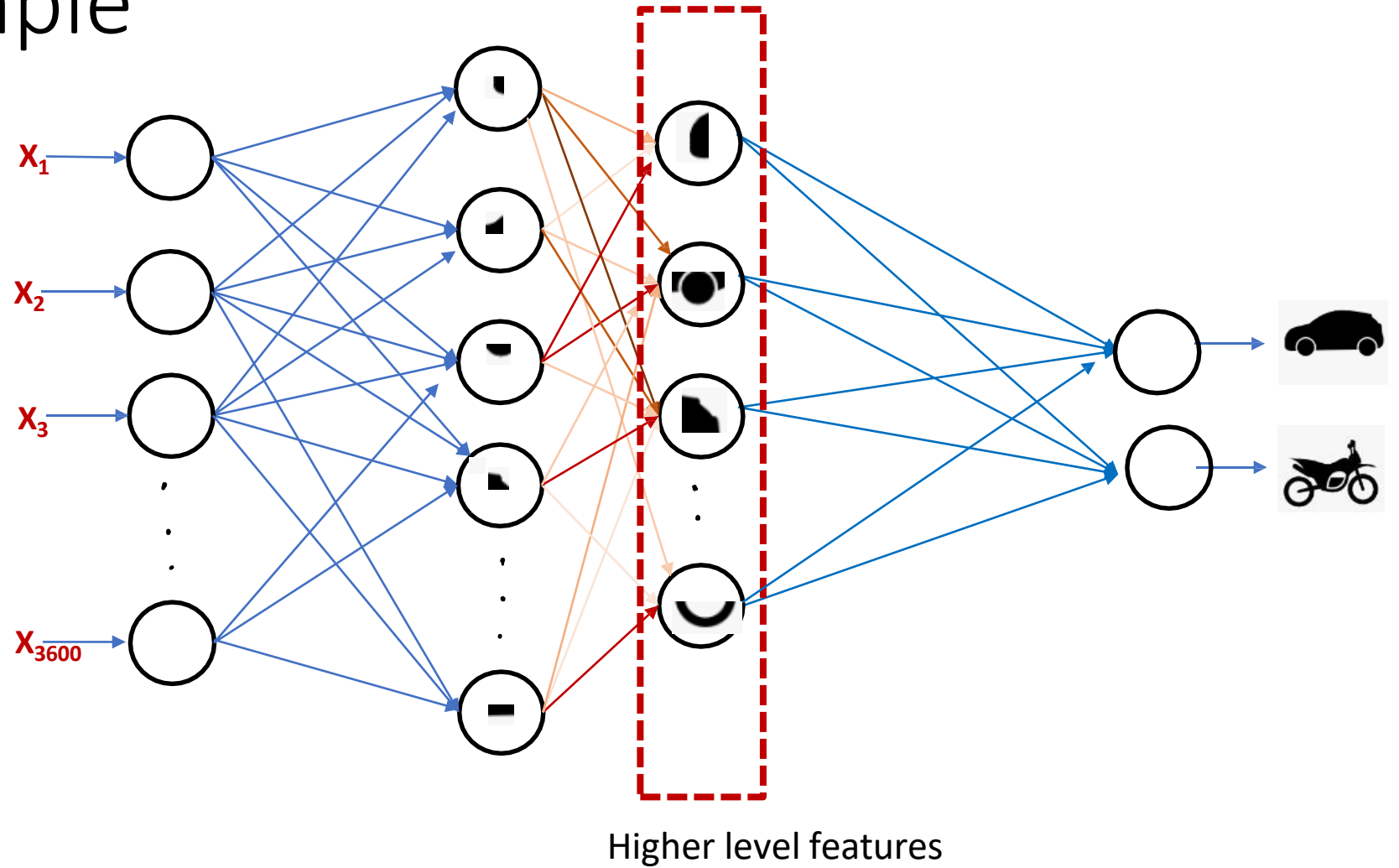


Lower level features

A Toy Example



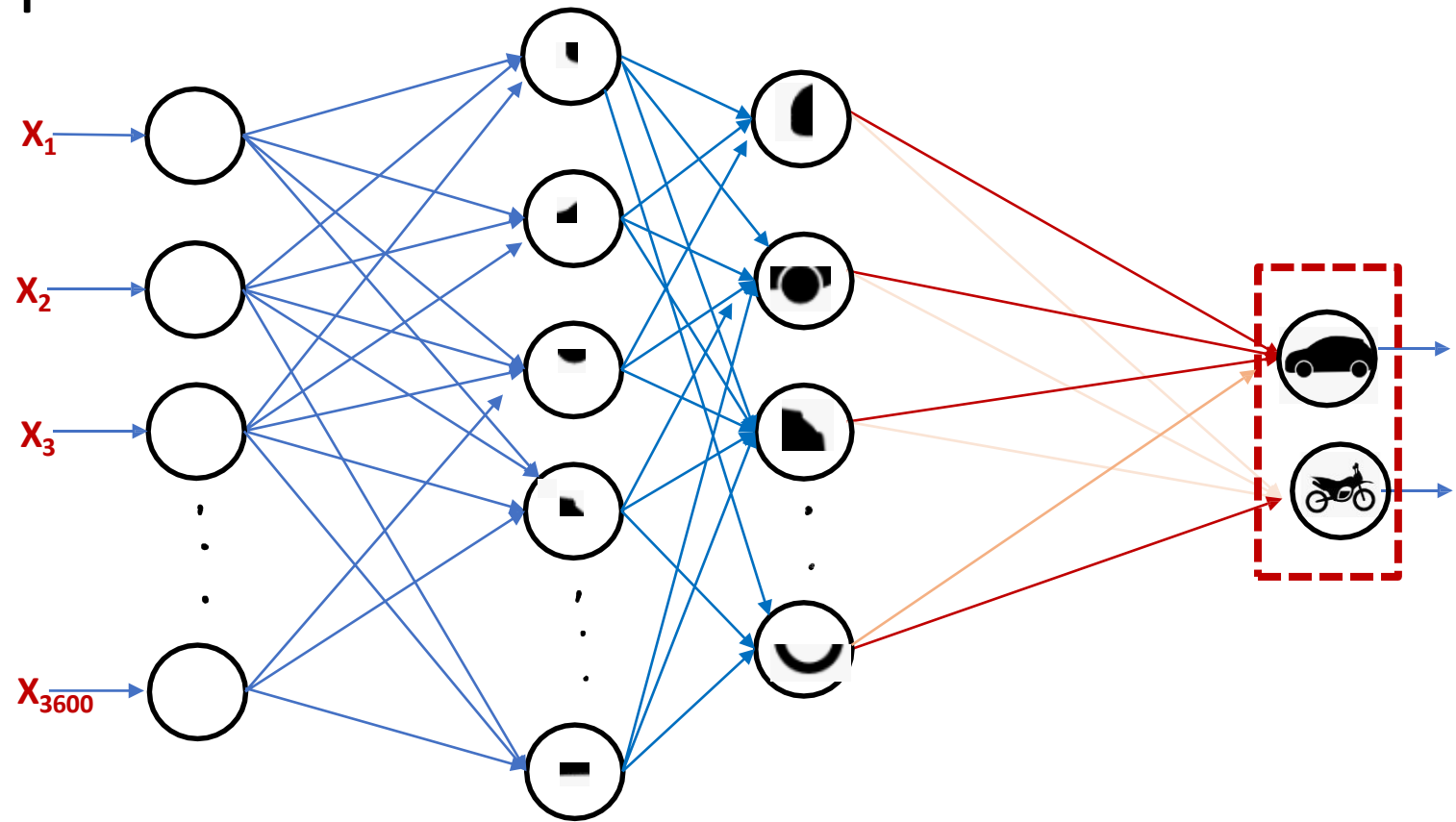
60x60=3600



A Toy Example



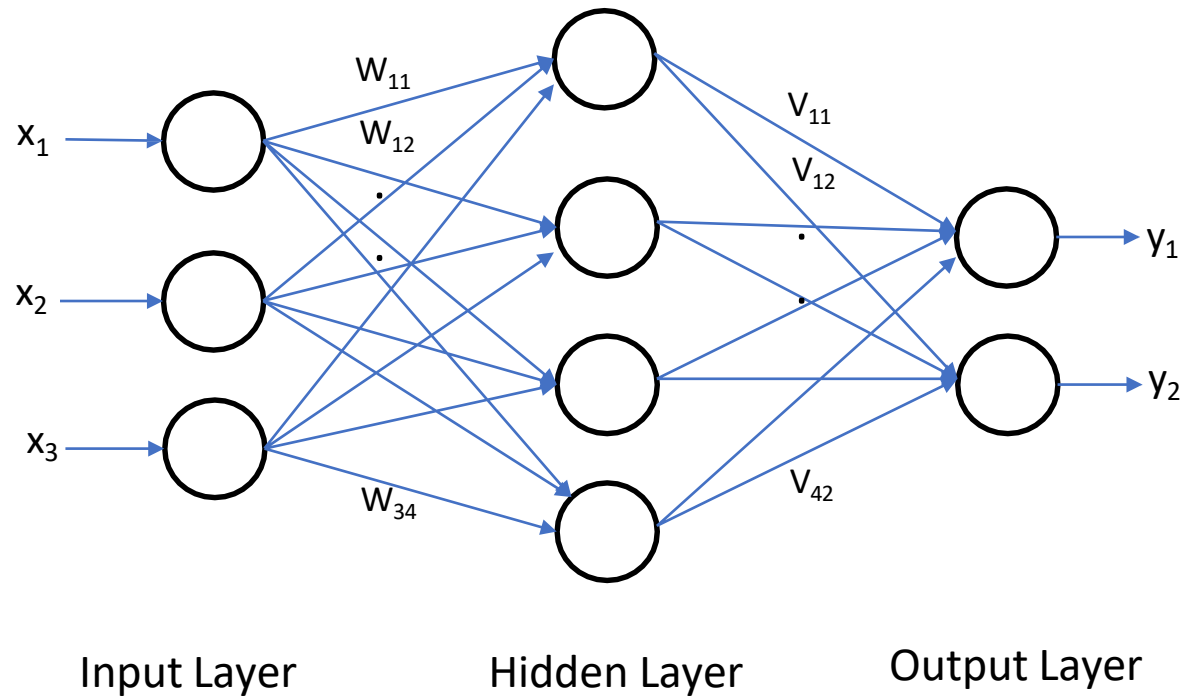
60x60=3600



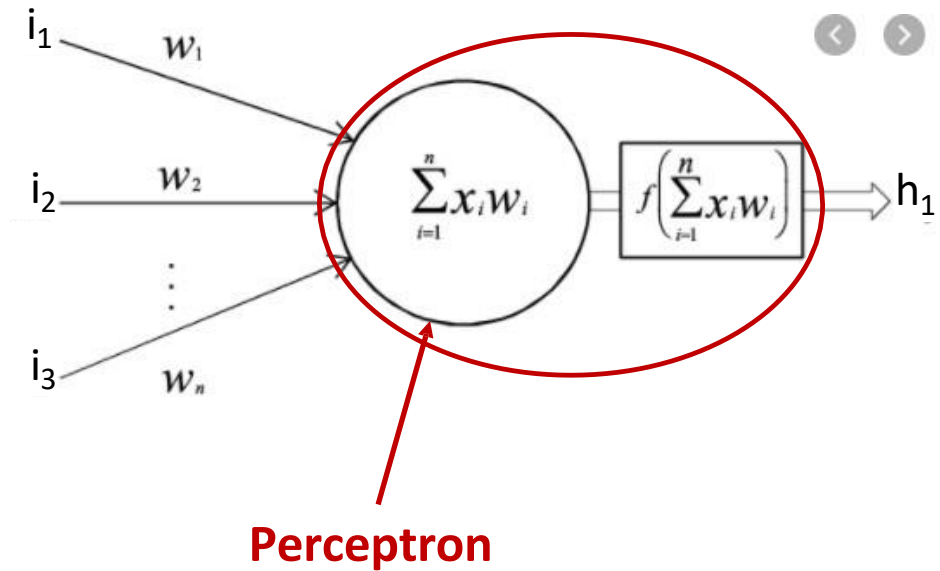
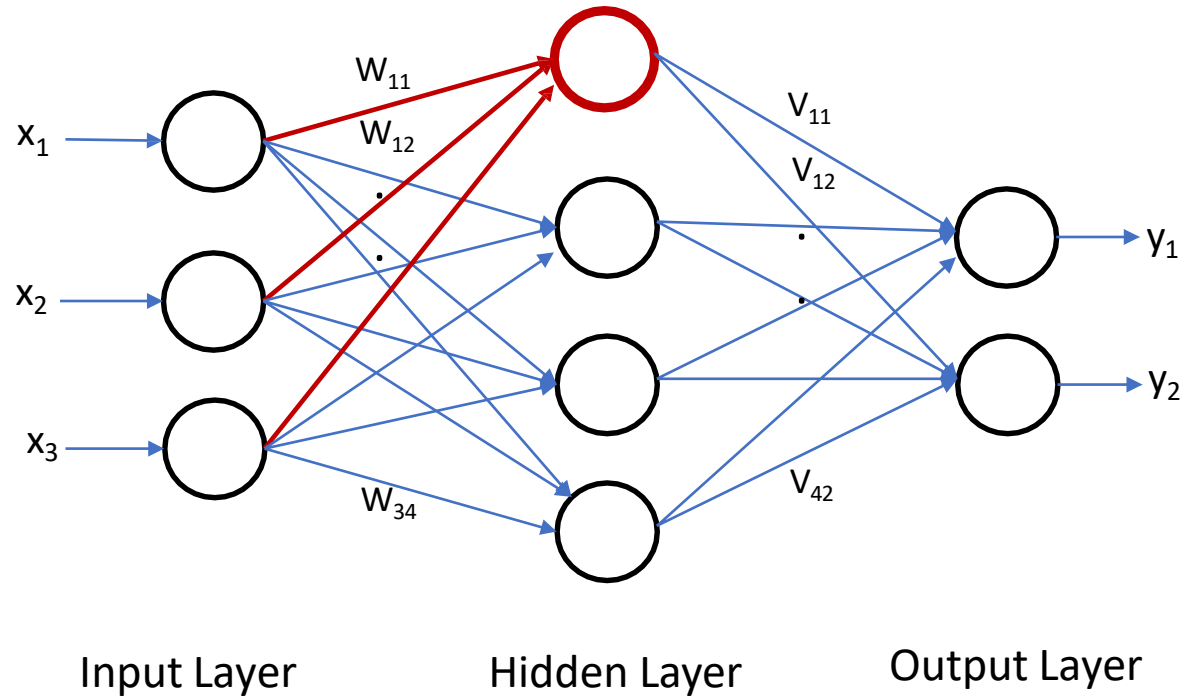
Higher level features

Multilayer Perceptron

What is Multilayer Perceptron Neural Network?



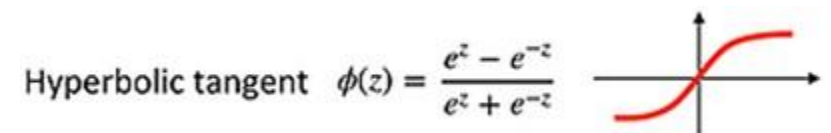
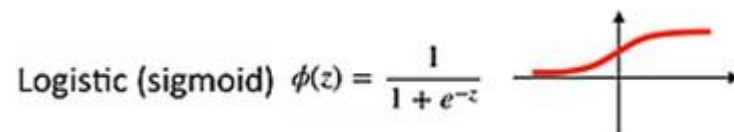
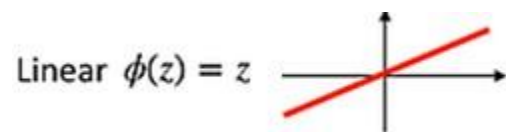
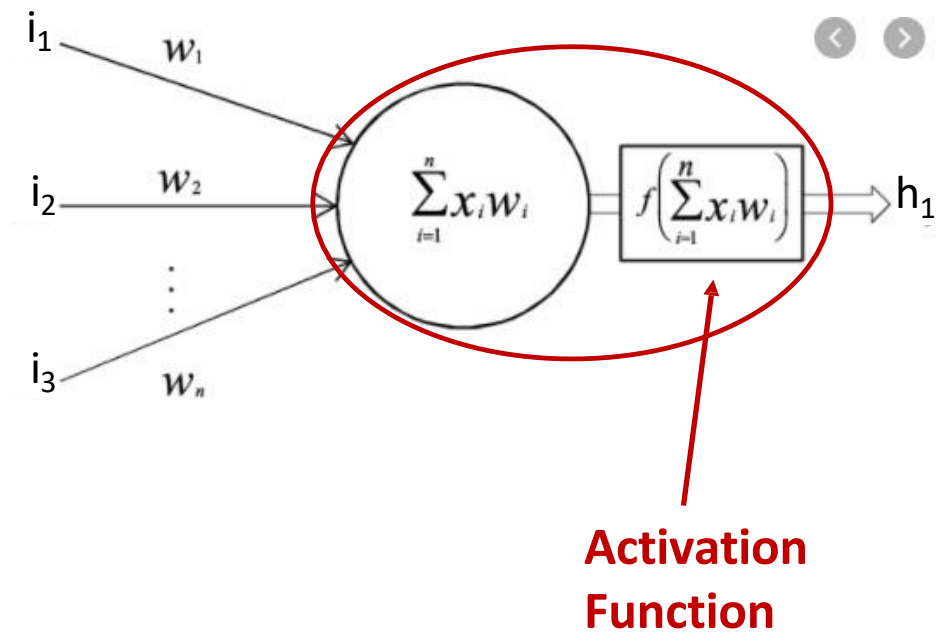
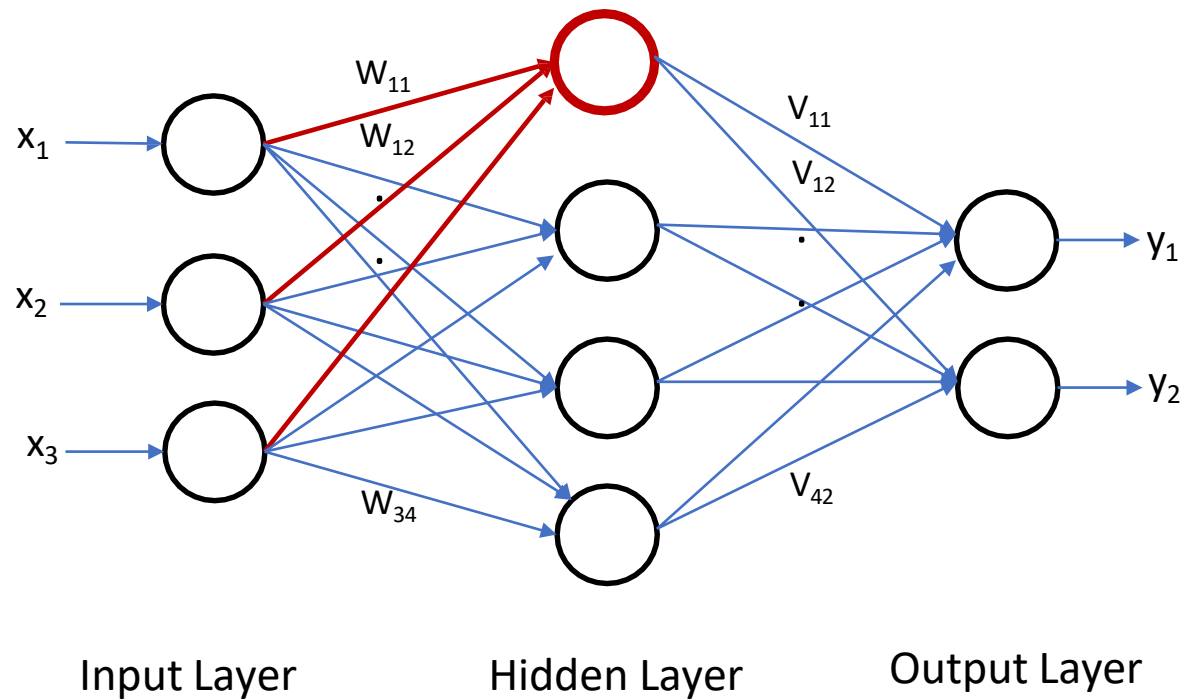
Multilayer Perceptron



For a given node i , the perceptron is defined as weighted summation of the incoming data from the nodes of the previous layer.

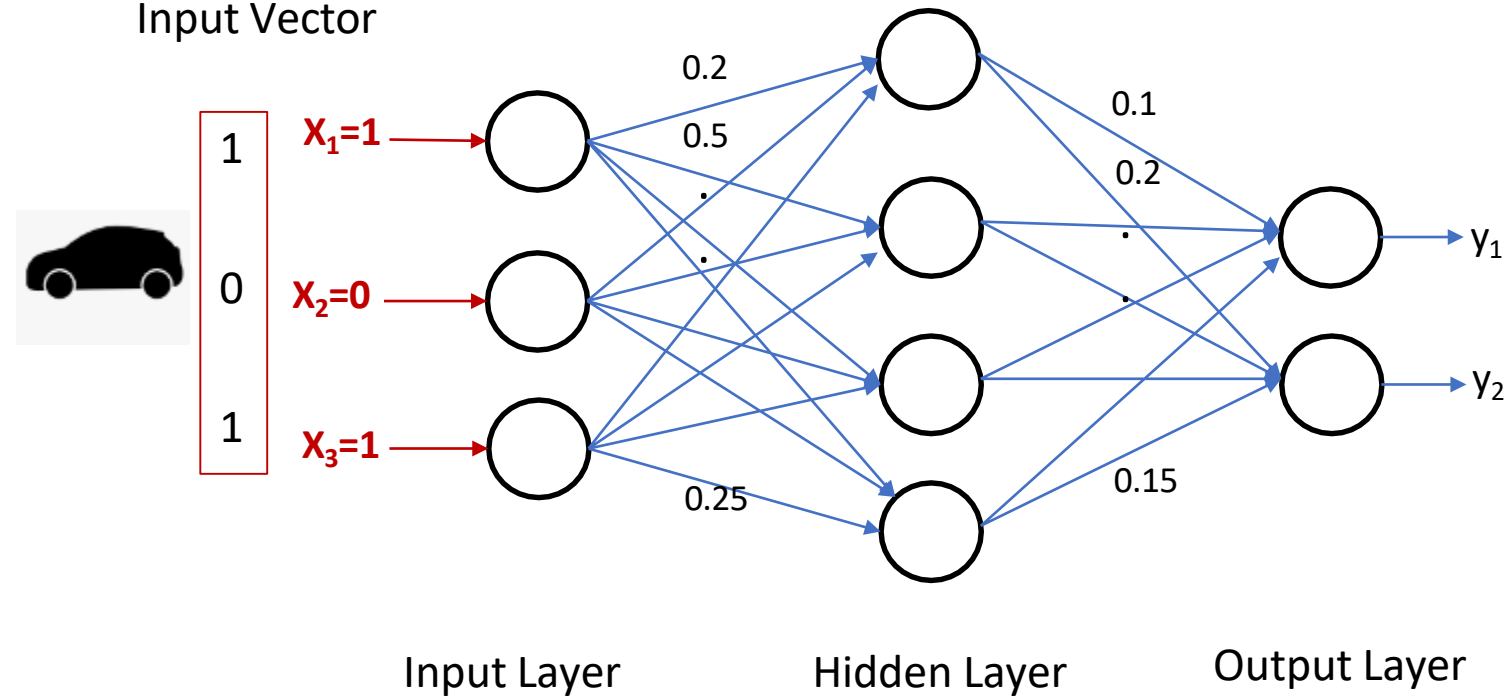
$$p_i = x_0 w_{0i} + x_2 w_{2i} + \dots + x_n w_{ni} = \sum_{j=1}^n x_j w_{ji}$$

Multilayer Perceptron

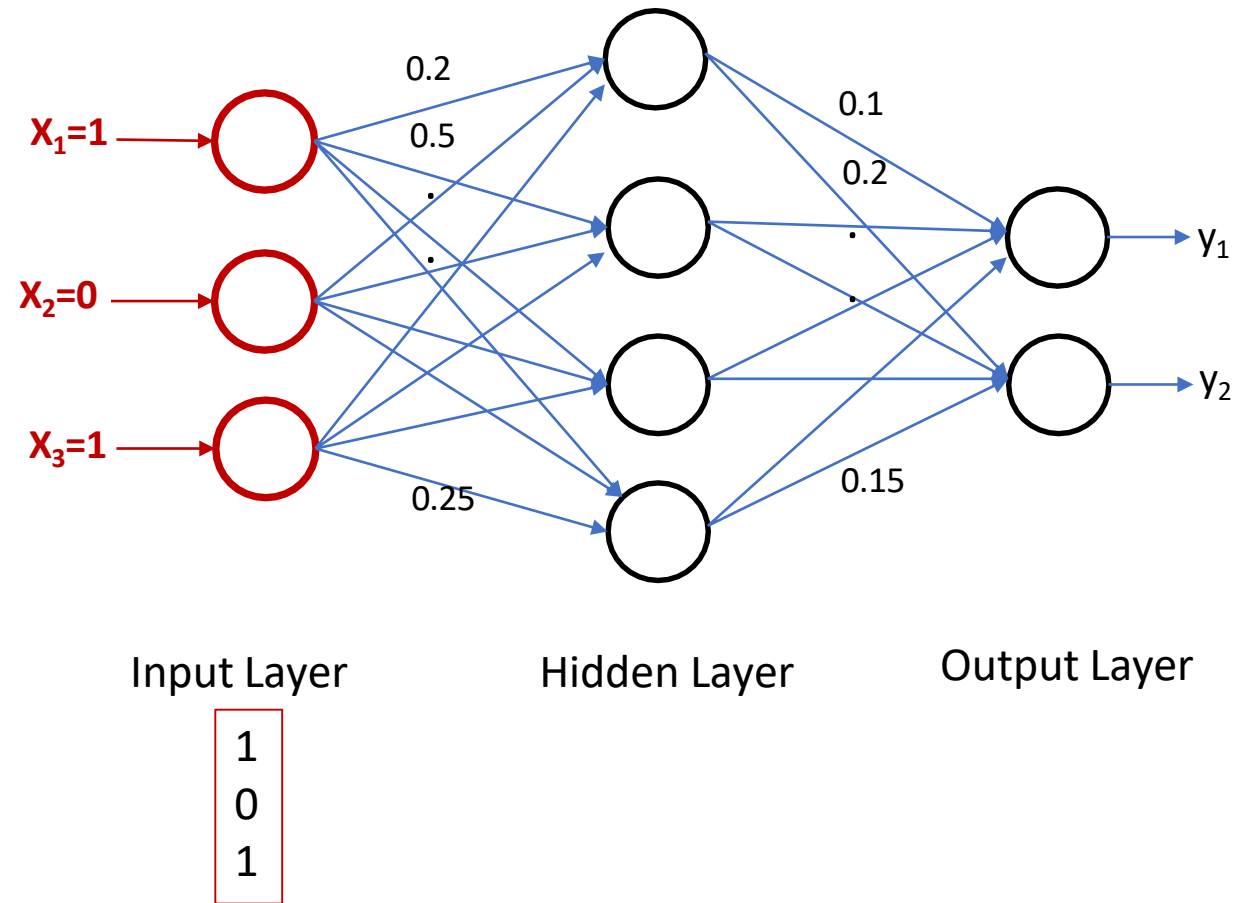


Forward Pass

Input Vector



Forward Pass

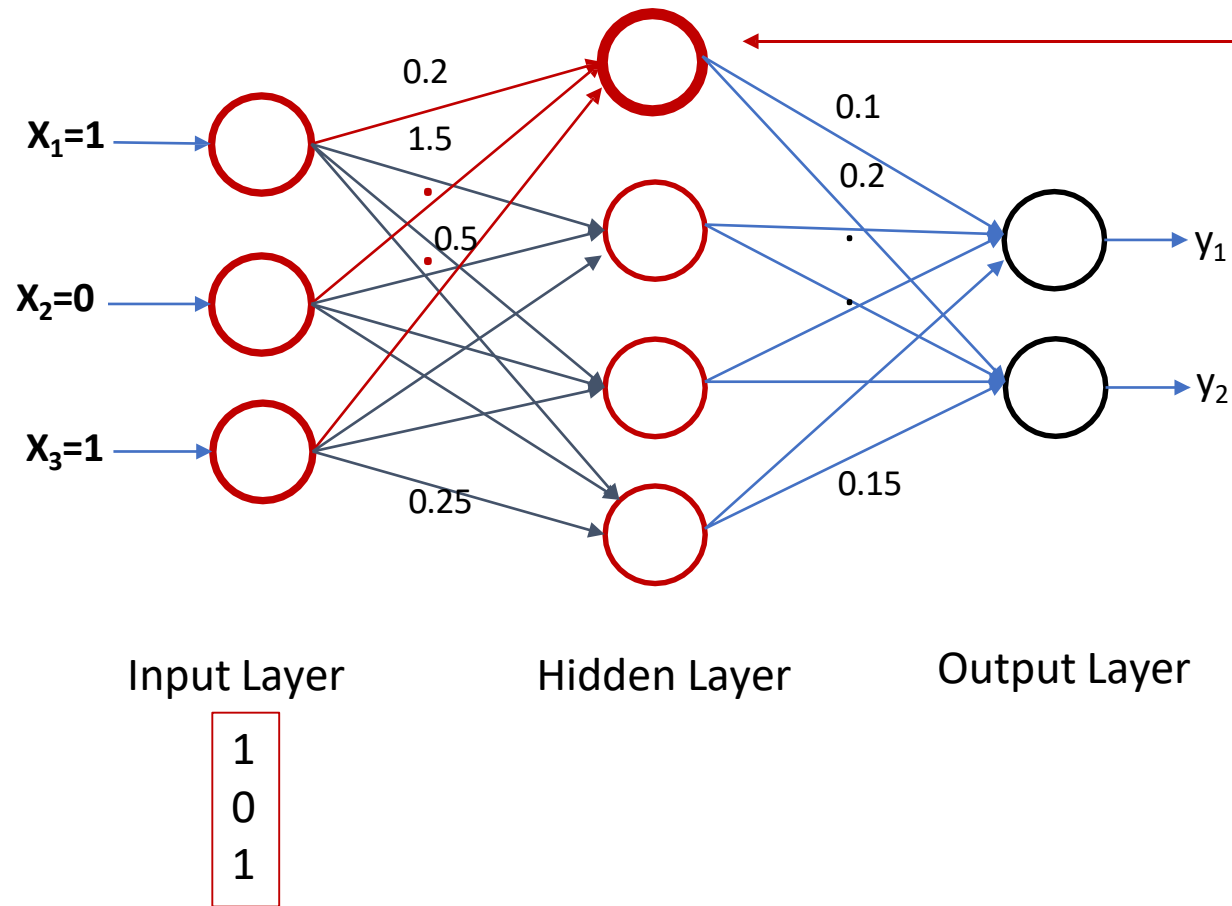


$$x = \text{Perceptron}(x)$$

Linear Activation function

$$x = f(x)$$

Forward Pass

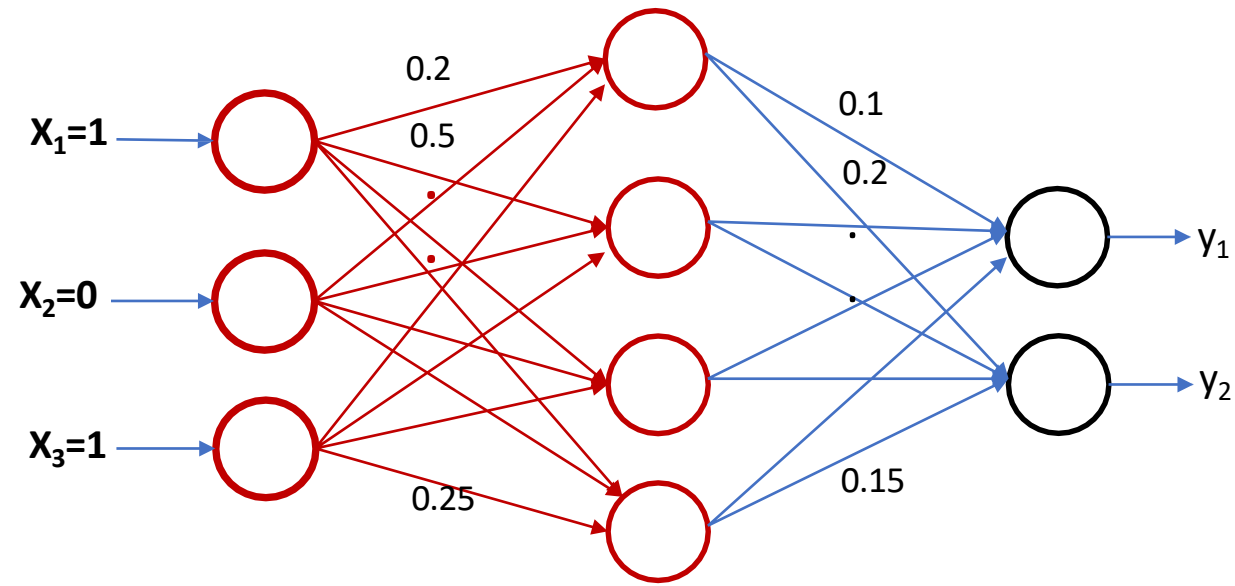


$$1 \times 0.2 + 0 \times 1.5 + 1 \times 0.5 = 0.7$$

Sigmoid Activation function

$$f(0.7) = 0.668$$

Forward Pass



Input Layer

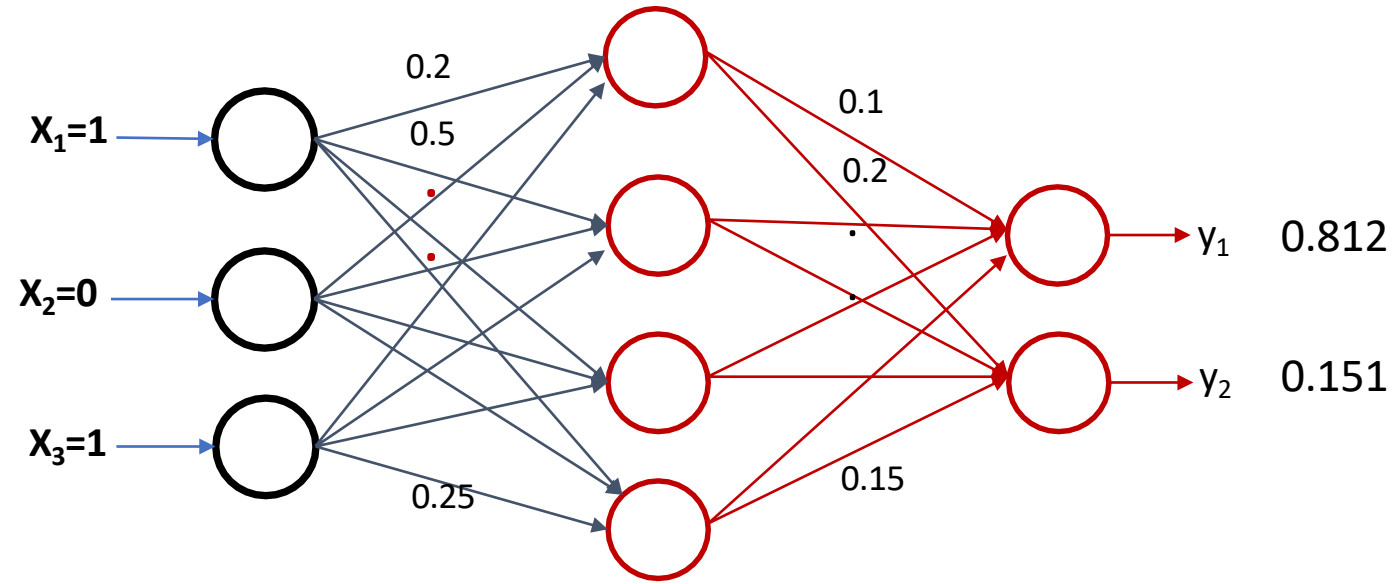
1
0
1

Hidden Layer

0.668
0.912
0.102
0.471

Output Layer

Forward Pass



Input Layer

1
0
1

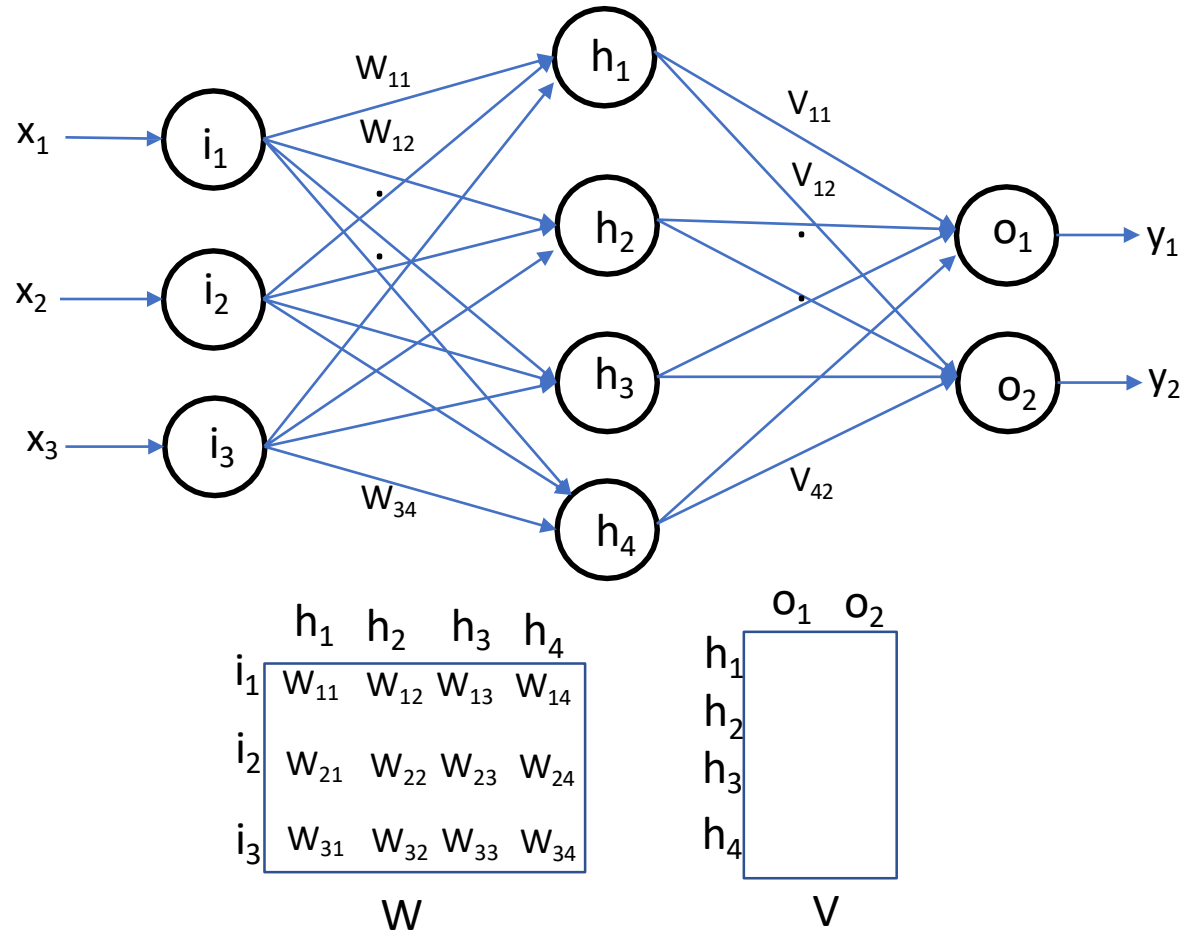
Hidden Layer

0.668
0.912
0.102
0.471

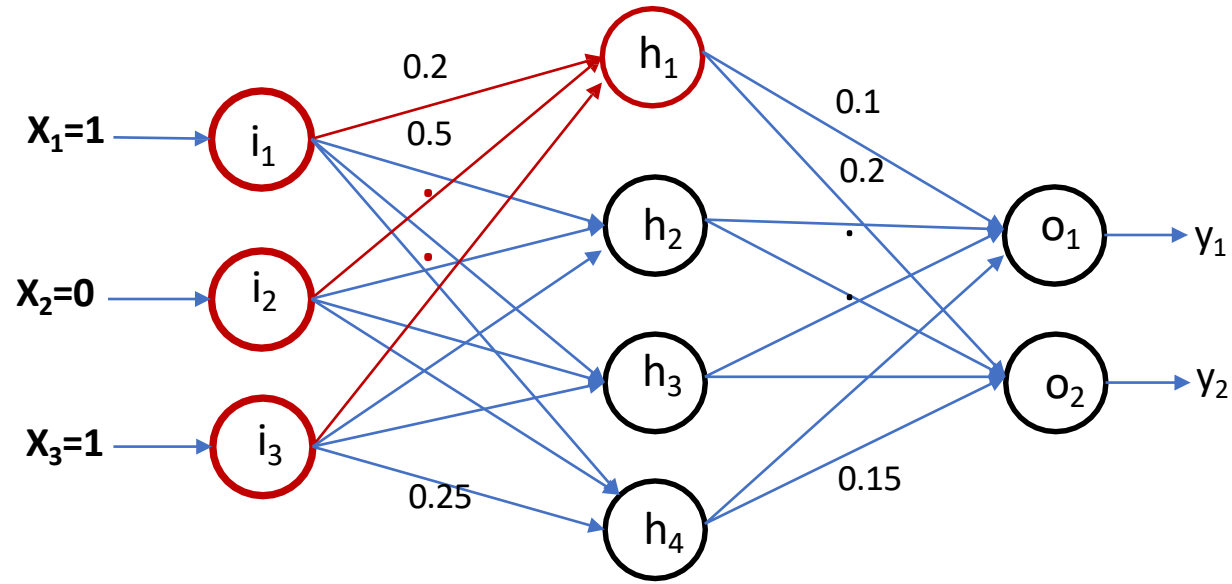
Output Layer

0.812
0.151

Weight Matrix



Forward Pass



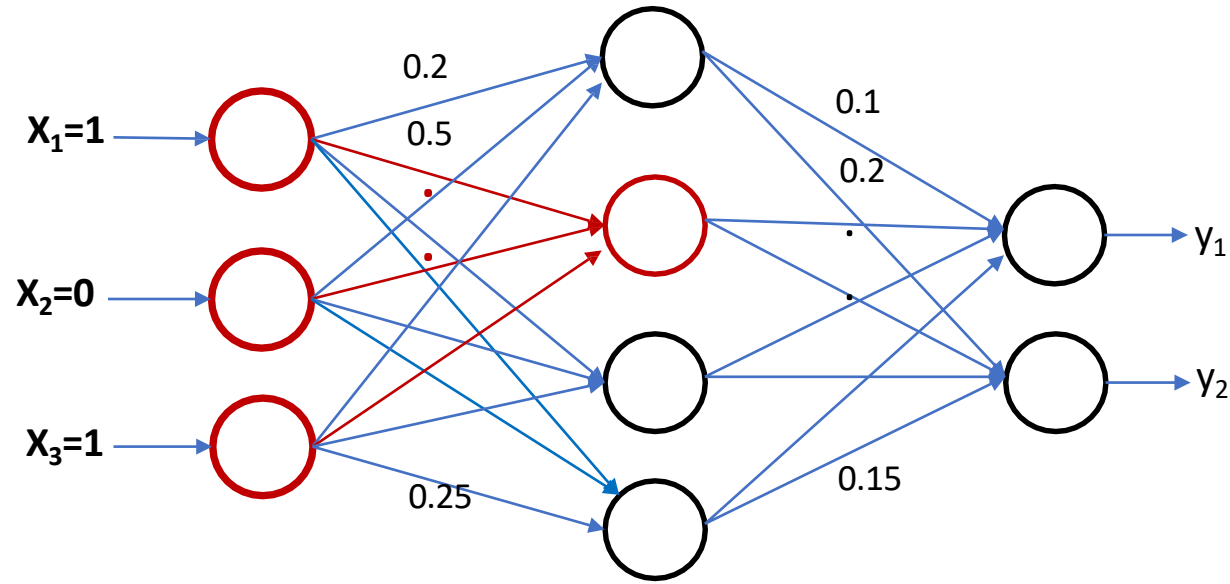
$$p_1 = x_1 W_{11} + x_2 W_{21} + x_3 W_{31}$$

$$h_1 = \text{Sigmoid}(p_1)$$

$$\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \times \begin{matrix} & \begin{matrix} h_1 & h_2 & h_3 & h_4 \end{matrix} \\ \begin{matrix} i_1 \\ i_2 \\ i_3 \end{matrix} & \begin{bmatrix} \boxed{0.2} & 0.5 & 0.1 & 0.25 \\ 0.1 & 0.2 & 0.1 & 0.15 \\ 0.25 & 0.1 & 0.1 & 0.15 \end{bmatrix} \end{matrix} = \begin{bmatrix} \boxed{0.668} & & & \end{bmatrix}$$

W

Forward Pass



$$p_2 = x_1W_{12} + x_2W_{22} + x_3W_{32}$$

$$h_2 = \text{Sigmoid}(p_2)$$

1	0	1
---	---	---

 $\times$

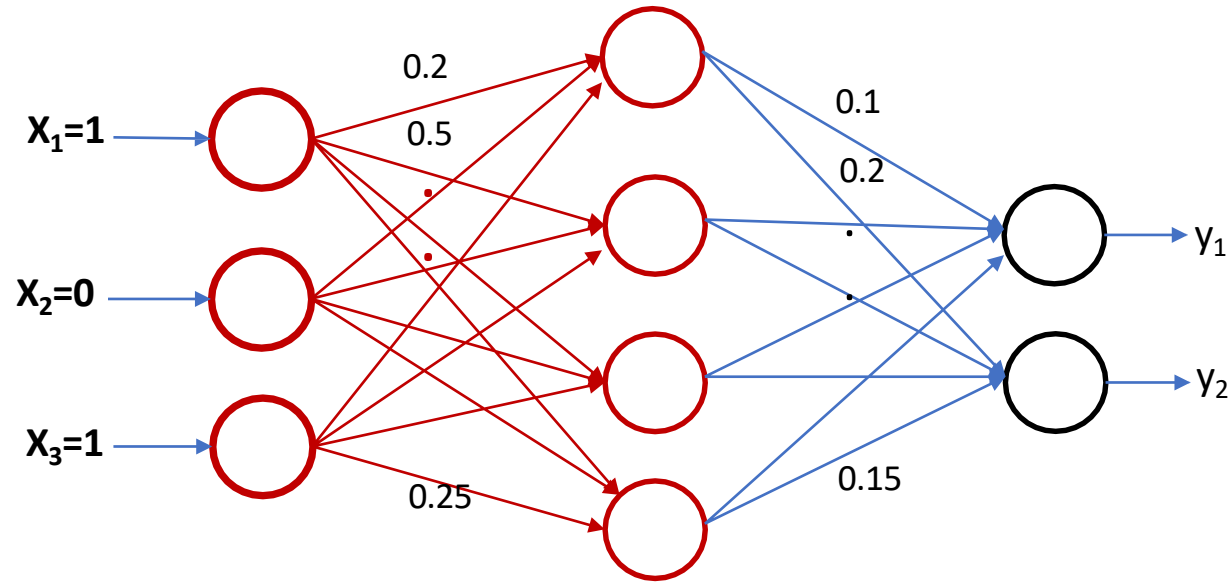
	h_1	h_2	h_3	h_4
i_1				
i_2				
i_3				

 $=$

h_1	h_2	h_3	h_4
0.668	0.912		

W

Forward Pass



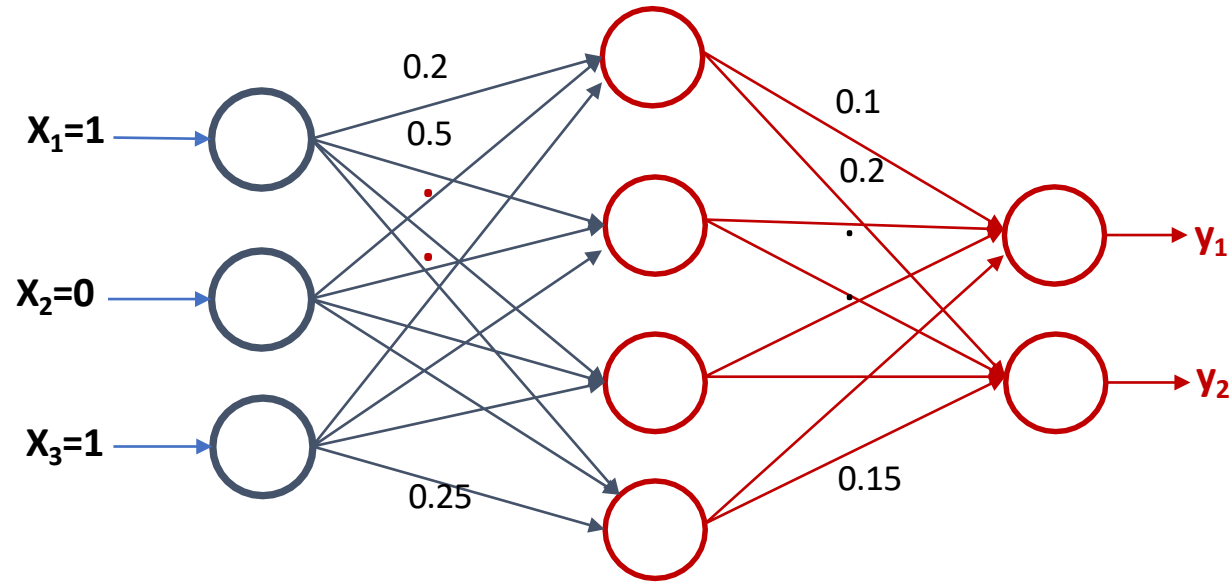
$$\begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline \end{array} \quad \mathbf{x} \quad \begin{array}{c} \begin{array}{cc} h_1 & h_2 & h_3 & h_4 \\ \hline \end{array} \\ \begin{array}{c} i_1 \\ i_2 \\ i_3 \end{array} \end{array} \quad \mathbf{W} = \begin{array}{|c|c|c|c|} \hline h_1 & h_2 & h_3 & h_4 \\ \hline 0.668 & 0.912 & 0.102 & 0.471 \\ \hline \end{array}$$

$\mathbf{x}$ $\mathbf{W}$ $\mathbf{h}$

$$\bar{p}^T = \bar{x}^T \cdot W$$

$$\bar{h}^T = \text{Sigmoid}(\bar{p}^T)$$

Forward Pass

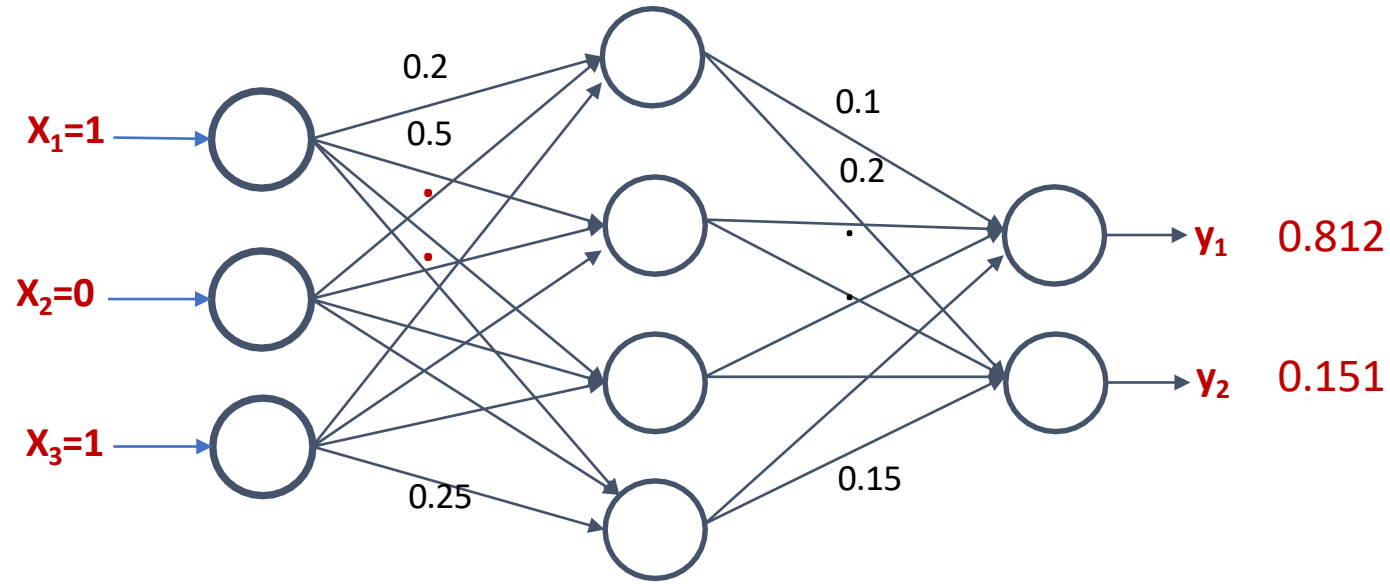


$$\begin{bmatrix} 0.668 & 0.912 & 0.102 & 0.471 \end{bmatrix} \times \begin{matrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{matrix} = \begin{bmatrix} 0.812 & 0.151 \end{bmatrix}$$

$\bar{h}^T$
 V
 $\bar{y}^T$

$$\begin{aligned}
 \bar{p}^T &= \bar{h}^T \cdot V \\
 \bar{y}^T &= \text{Sigmoid}(\bar{p}^T)
 \end{aligned}$$

Forward Pass



$$\bar{y}^T = \text{Sigmoid}(\text{Sigmoid}(\bar{x}^T W)^T \cdot V)$$